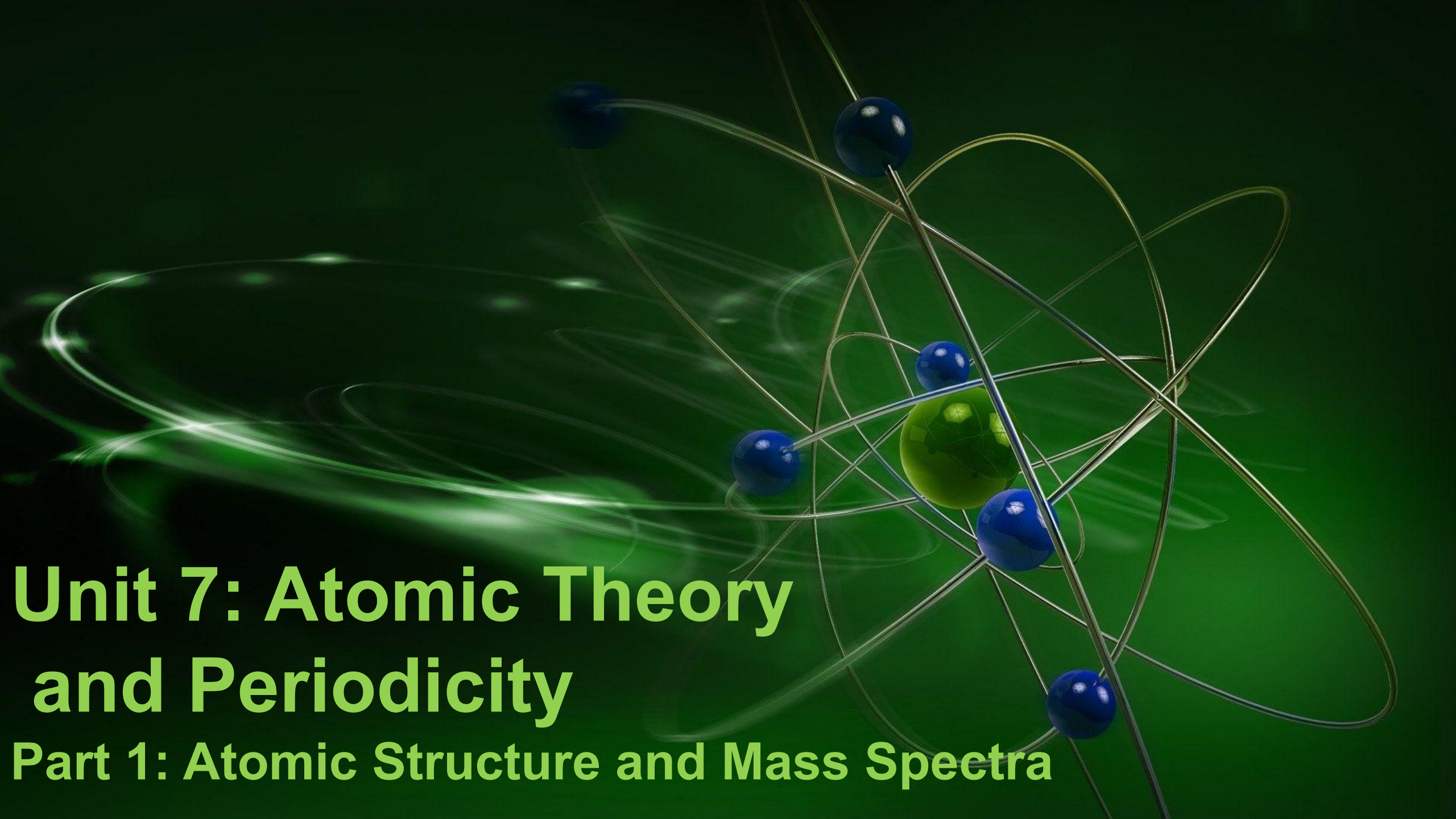


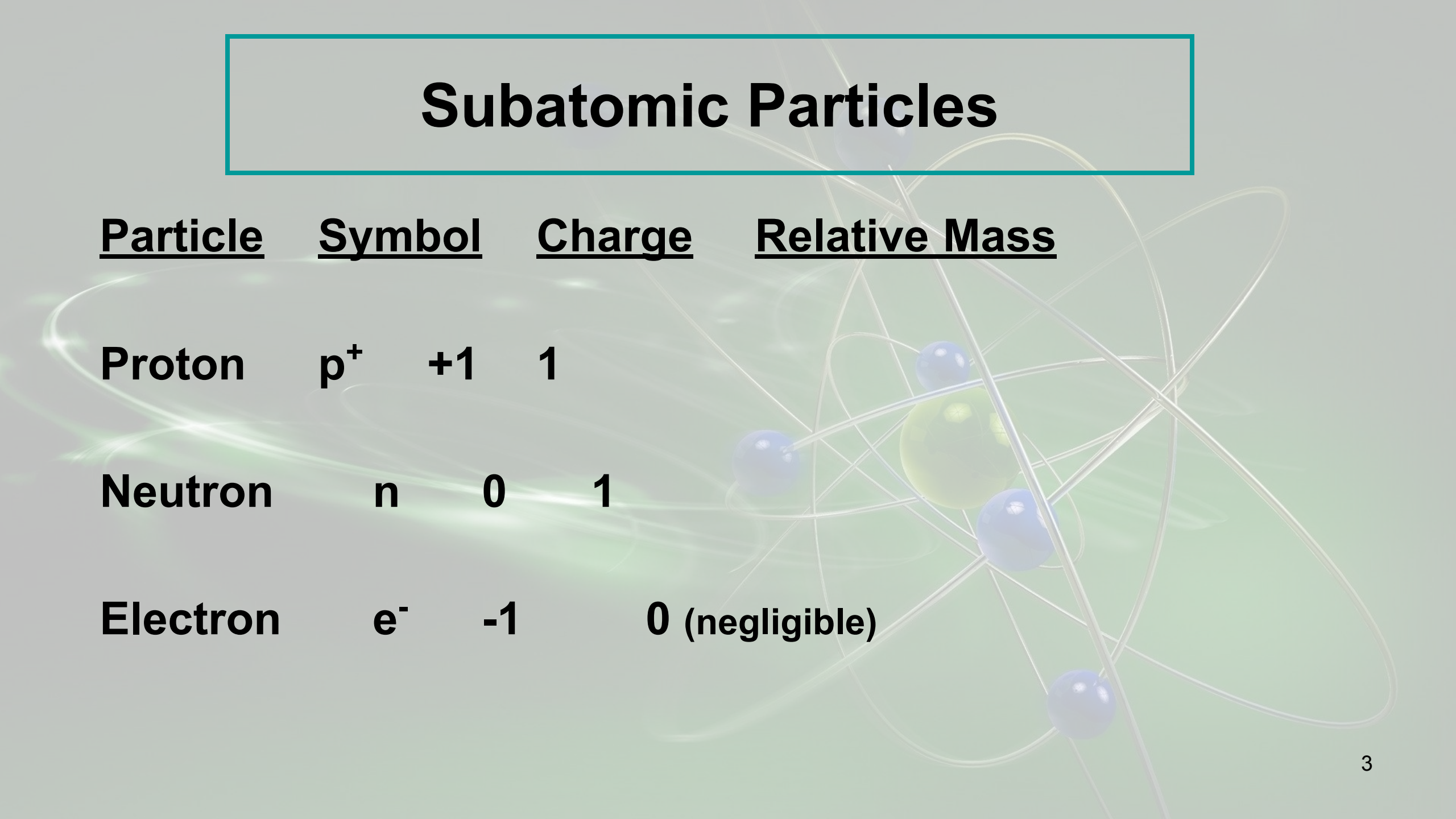
**Unit 8: Atomic Theory and
Periodicity
Zumdahl Chapter 7**



Unit 7: Atomic Theory and Periodicity

Part 1: Atomic Structure and Mass Spectra

Subatomic Particles



<u>Particle</u>	<u>Symbol</u>	<u>Charge</u>	<u>Relative Mass</u>
Proton	p^+	+1	1
Neutron	n	0	1
Electron	e^-	-1	0 (negligible)

Isotopes

- Atoms with the same number of protons, but different numbers of neutrons.
- Atoms of the same element (same atomic number) with different mass numbers

Isotopes of chlorine

^{35}Cl

^{37}Cl

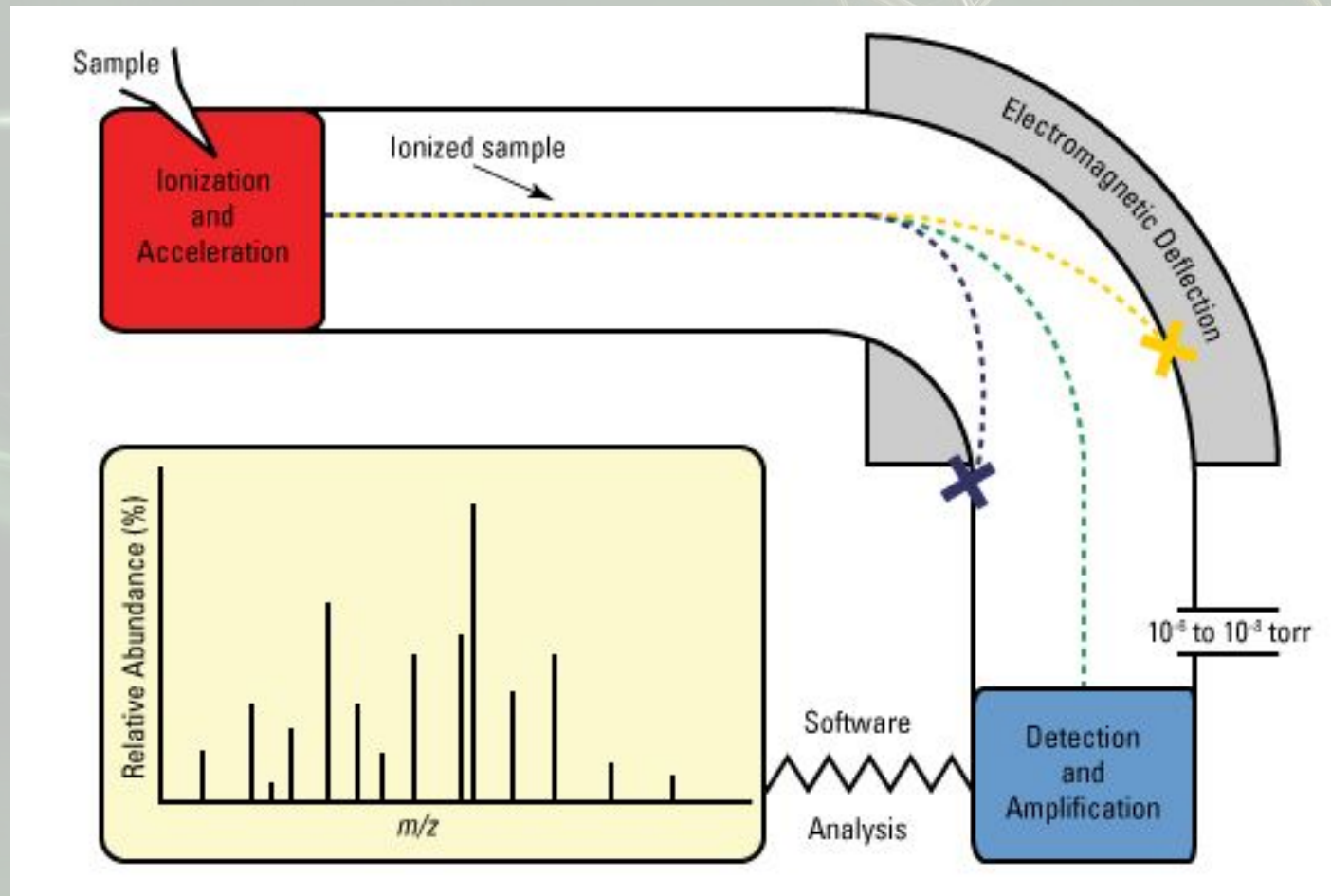
17

17

chlorine - 35

chlorine - 37

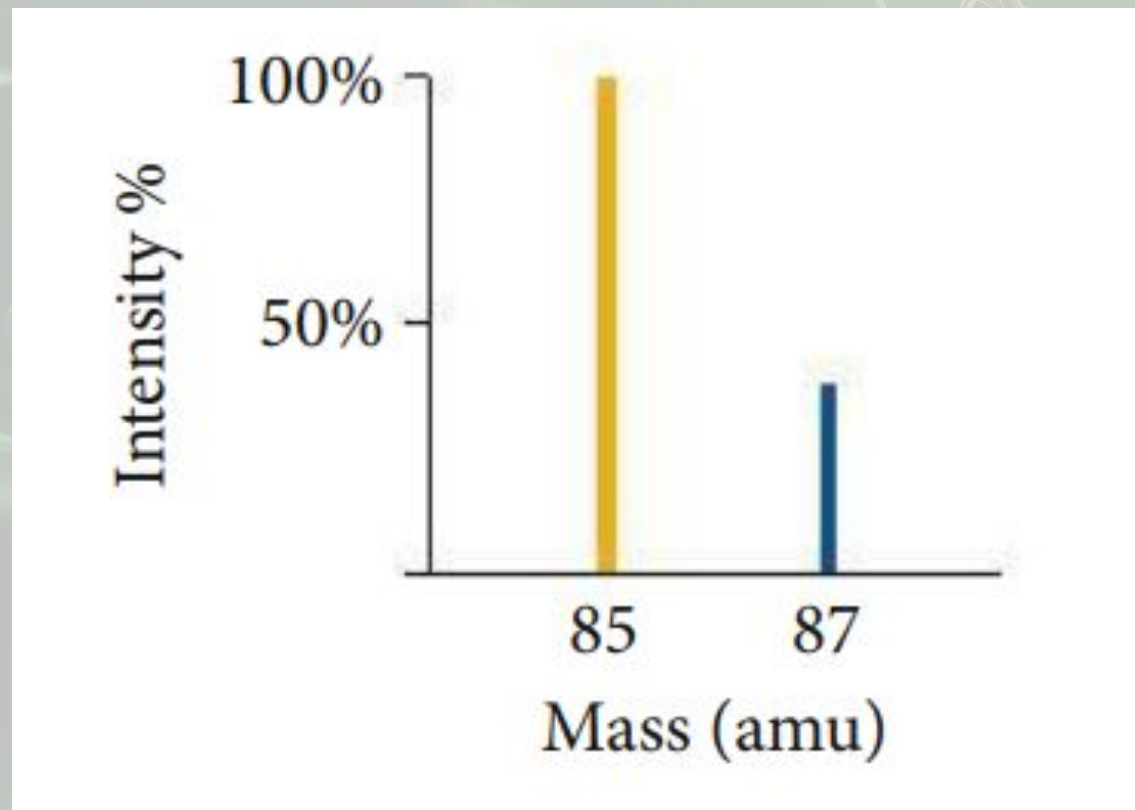
Mass Spectrometry



Weighted Average Problems

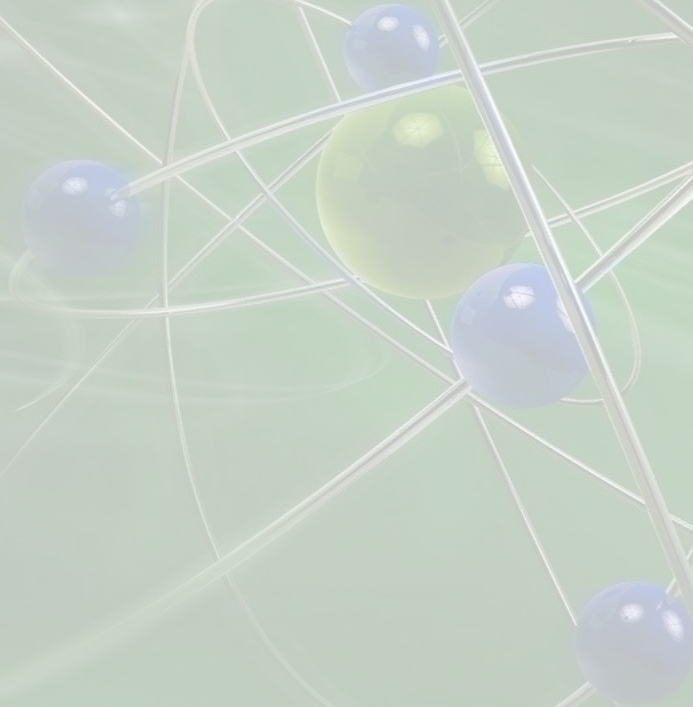
- The isotopes of chlorine are Cl-35 and Cl-37. In a mass spectrometer analysis, 160 chlorine particles were analyzed, and 124 were found to have mass 34.998, and the remainder mass 36.973. What is the average atomic mass?

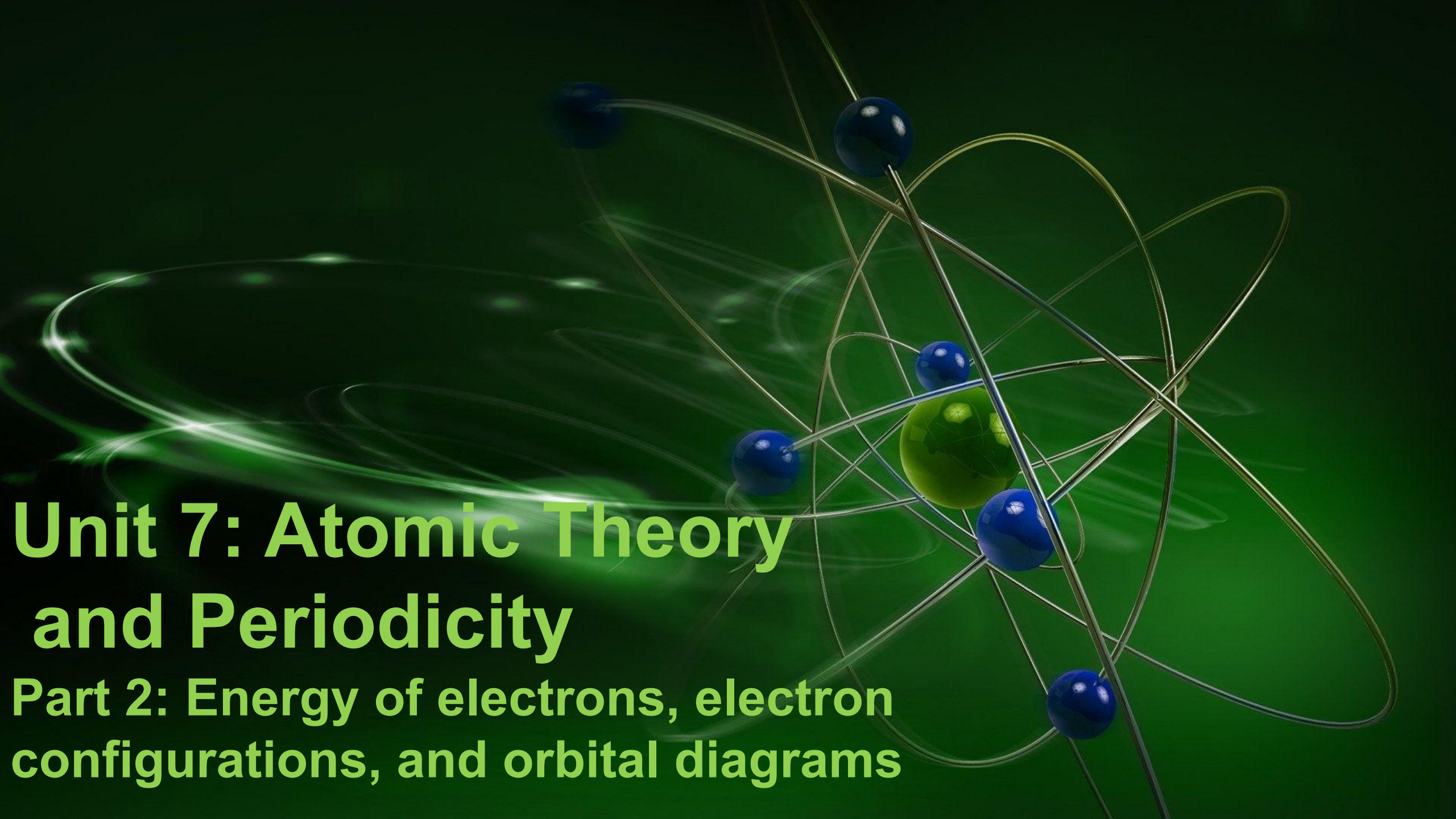
Calculate an average atomic mass and identify the atom



Whiteboard time-

- Draw the mass spectra of a sample of chlorine gas. Label the axis and represent the correct peak ratios.





Unit 7: Atomic Theory and Periodicity

**Part 2: Energy of electrons, electron
configurations, and orbital diagrams**

The Electromagnetic Spectrum

The electromagnetic spectrum represents the range of light energy

low energy

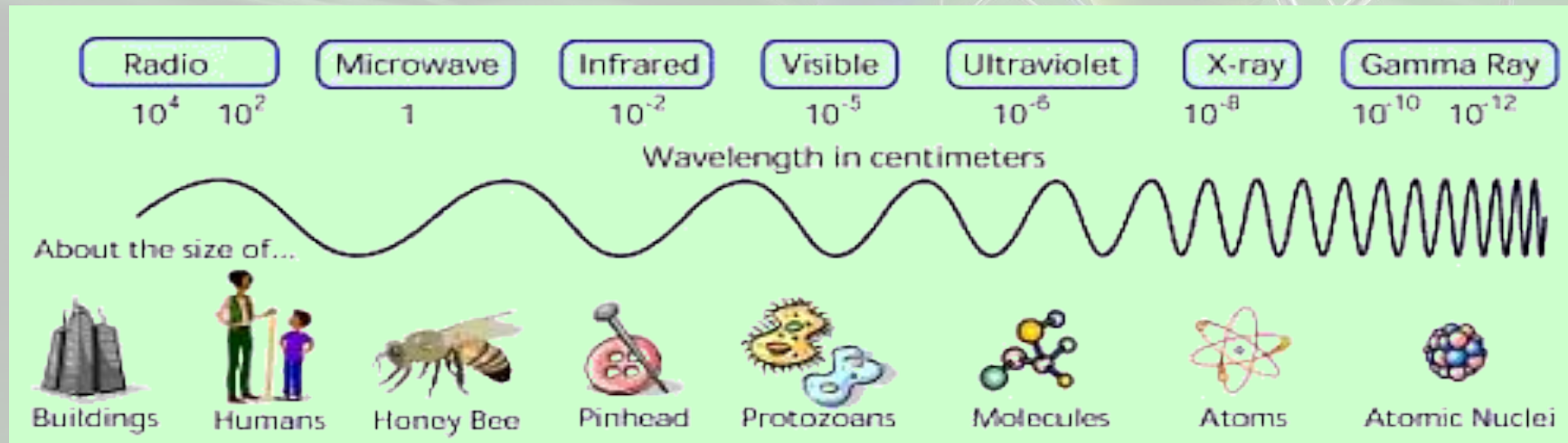
high energy

low frequency

high frequency

long wavelengths

short wavelengths



Visible light is a small portion of this spectrum that our eyes can detect. What we see is a rainbow of colors. All the colors together create white light.

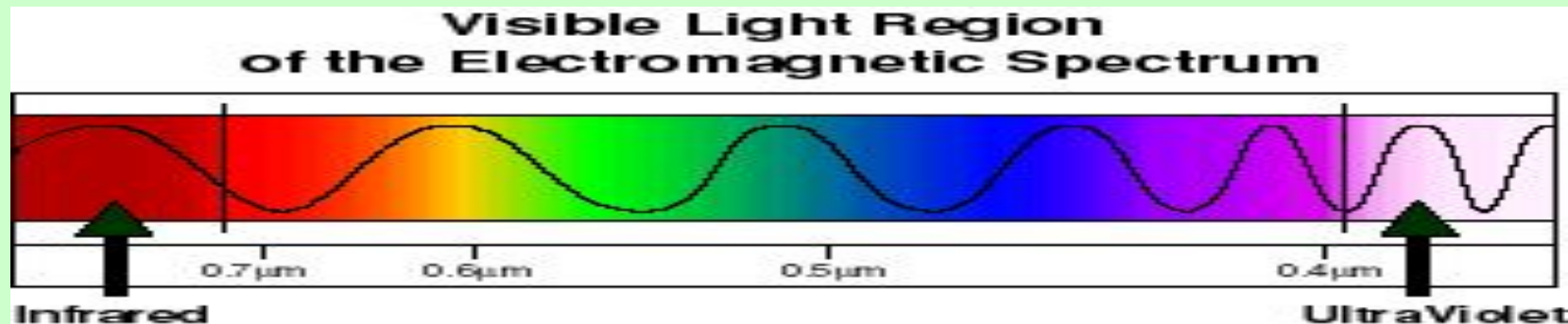
Red Orange Yellow Green Blue Indigo Violet

Low energy



high energy

ROY G BIV



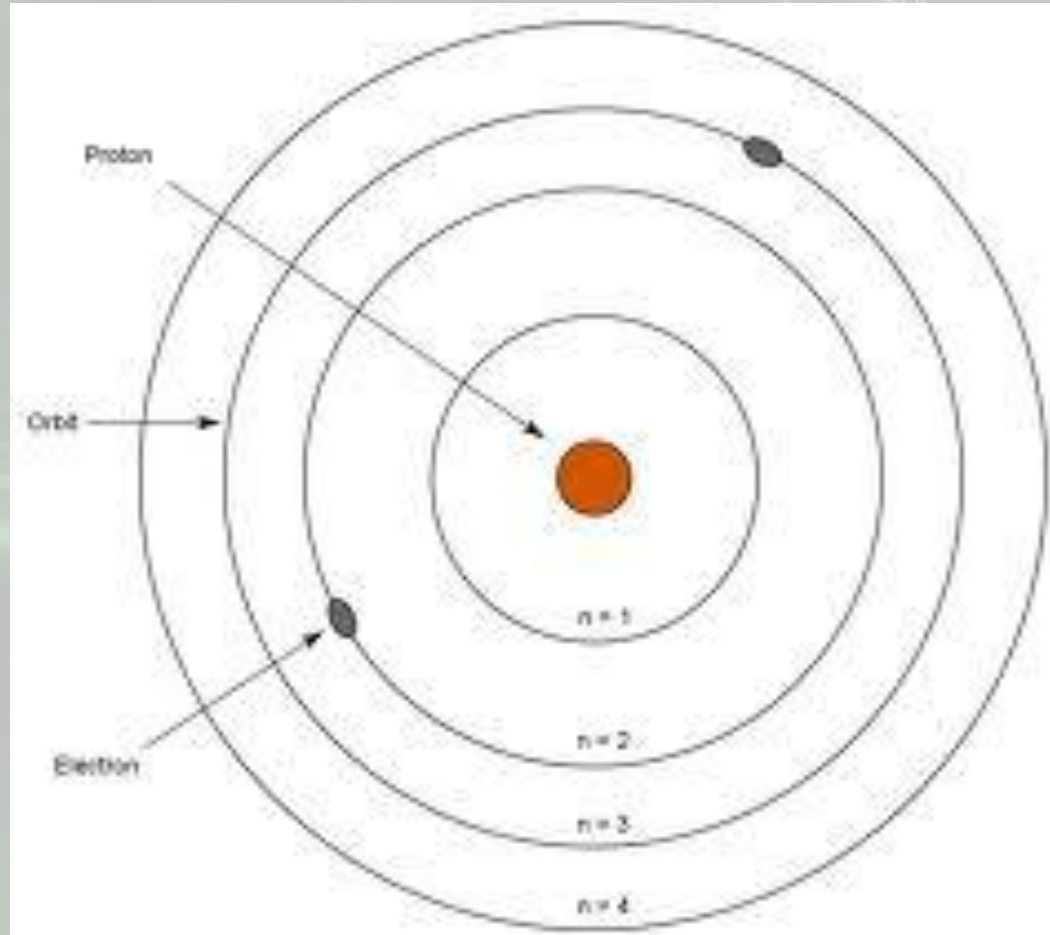
$$c = \lambda \nu$$

- The frequency (ν) of a wave is the number of waves to cross a point in 1 second (units are Hertz – cycles/sec or sec^{-1})
- λ is the wavelength- the distance from crest to crest on a wave
- $c = 3.00 \times 10^8 \text{ m/s}$, speed of light
- $c = 2.998 \times 10^8 \text{ m/s}$, speed of light?

$$E = h\nu$$

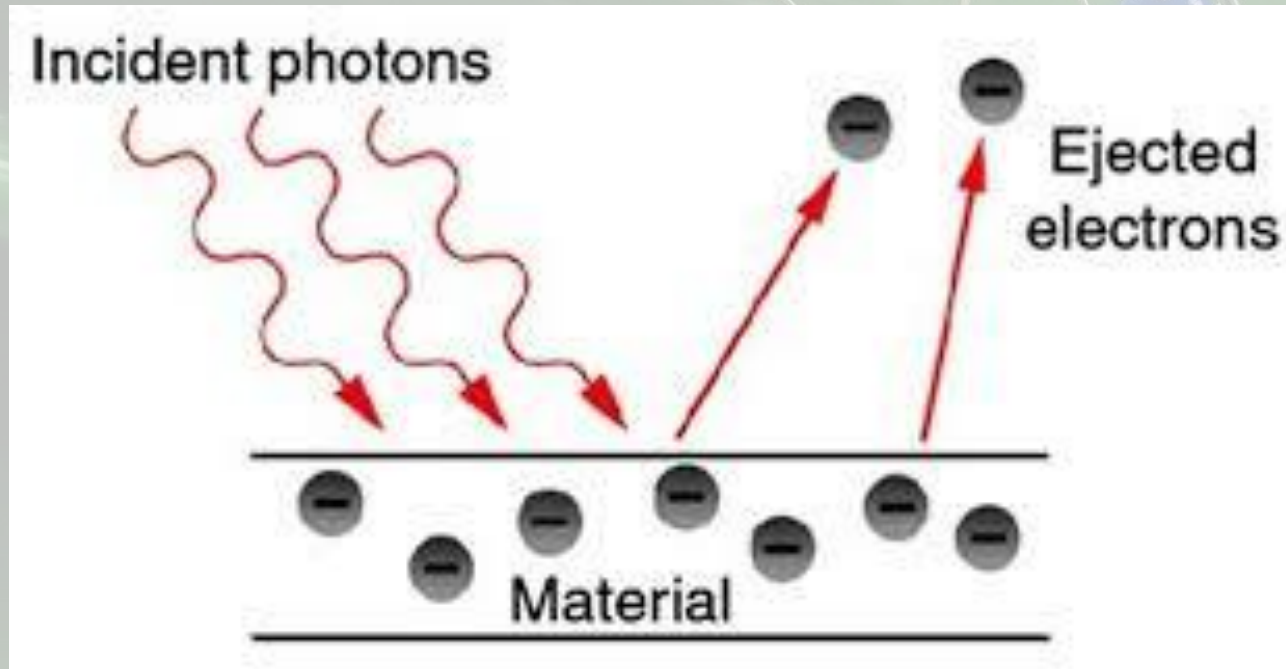
- ν is still the frequency (Hertz or s^{-1})
- E is the energy of the photon (Joules)
- $h = 6.626 \times 10^{-34} \text{ Js}$, Planck's constant

Back to Light and Atoms



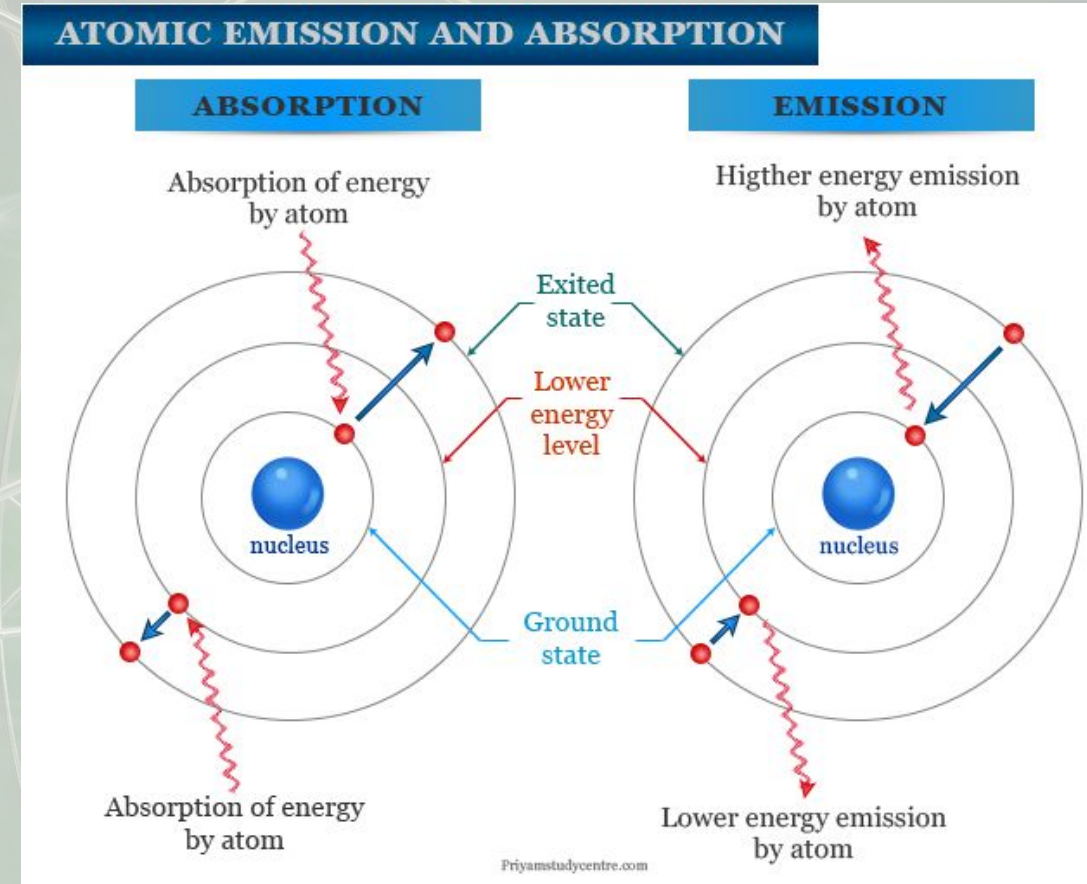
The Photoelectric Effect

Shine high energy light on a material,
and electrons will be ejected from it!



Atoms and Light

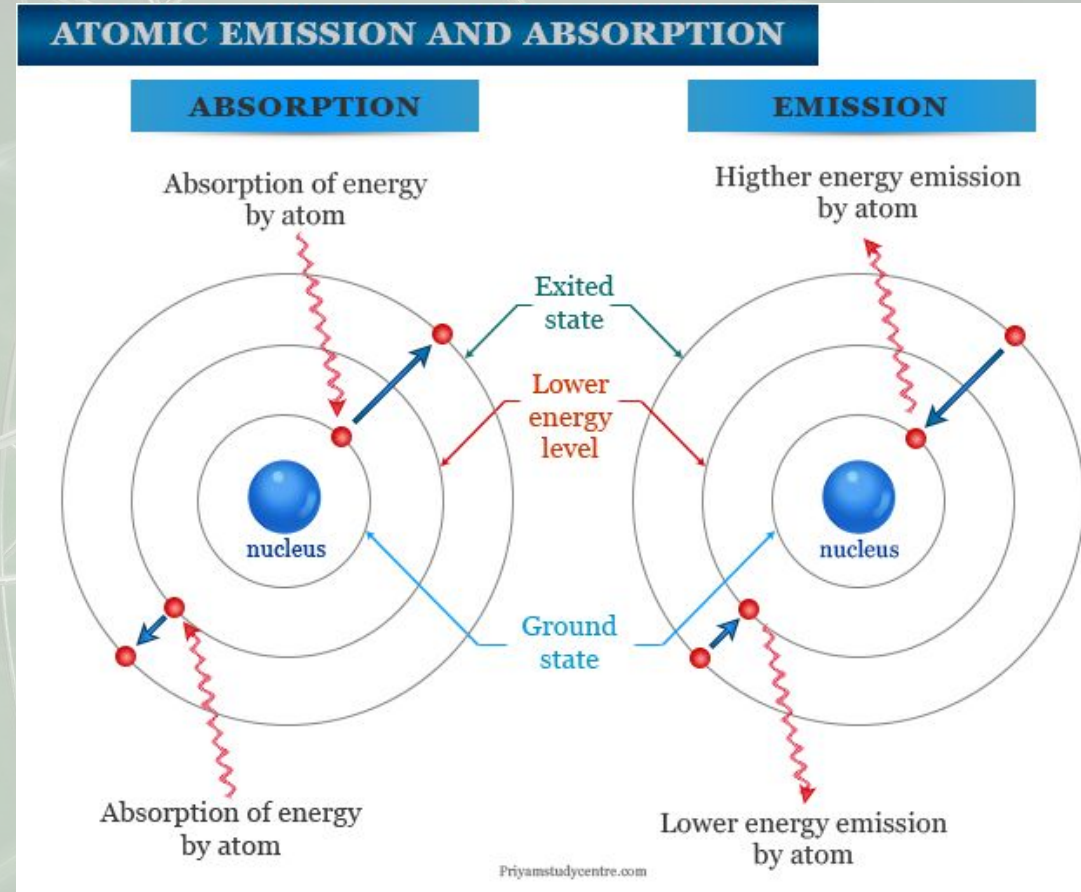
- Electrons in atoms can **absorb** light energy- sometimes enough to free them! (Photoelectric effect!)
- Sometimes it is enough to take an electron from its **ground state** (least energetic) to an **excited state** (energized!)
- Each element absorbs only certain frequencies of light, called the **absorption spectrum**.



Emission Spectra

- Once the electrons are in the excited state they can '**relax**' back to the ground state by **emitting** light
- This is called the '**emission spectrum**' and it is different for each element. Emission spectra can therefore be used to **identify** types of element
- The light **absorbed or emitted** has an energy, frequency, and wavelength found with:

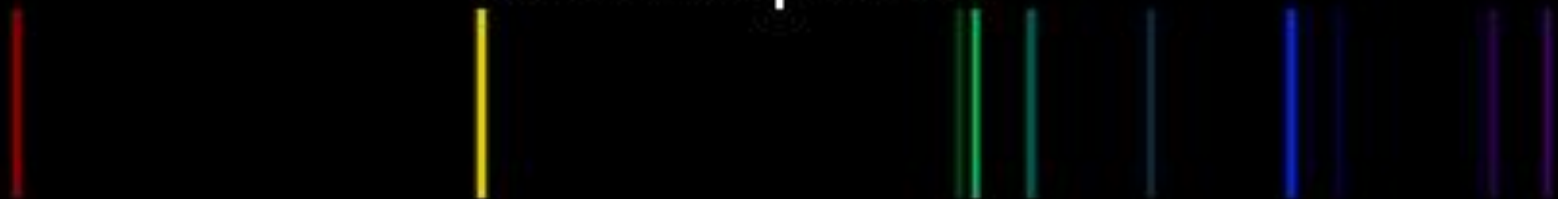
$$c = \lambda \nu \text{ and } E = h\nu$$



Continuous Spectrum



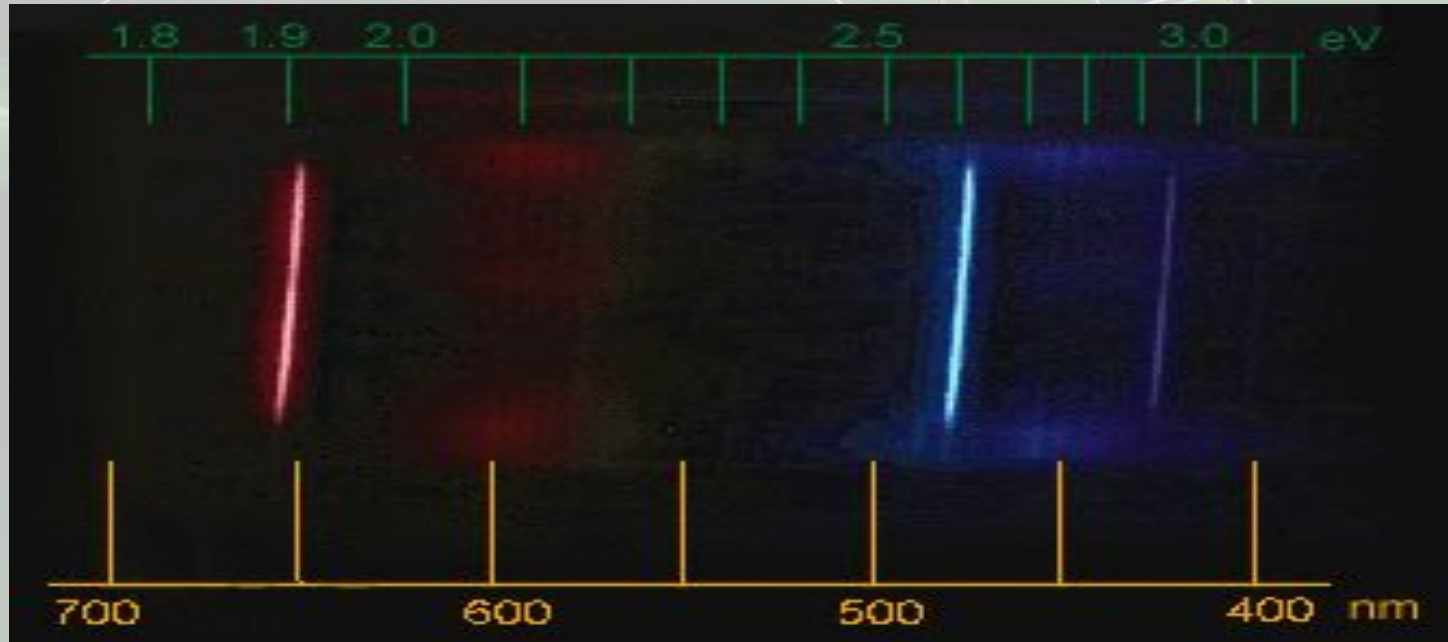
Emission Spectrum



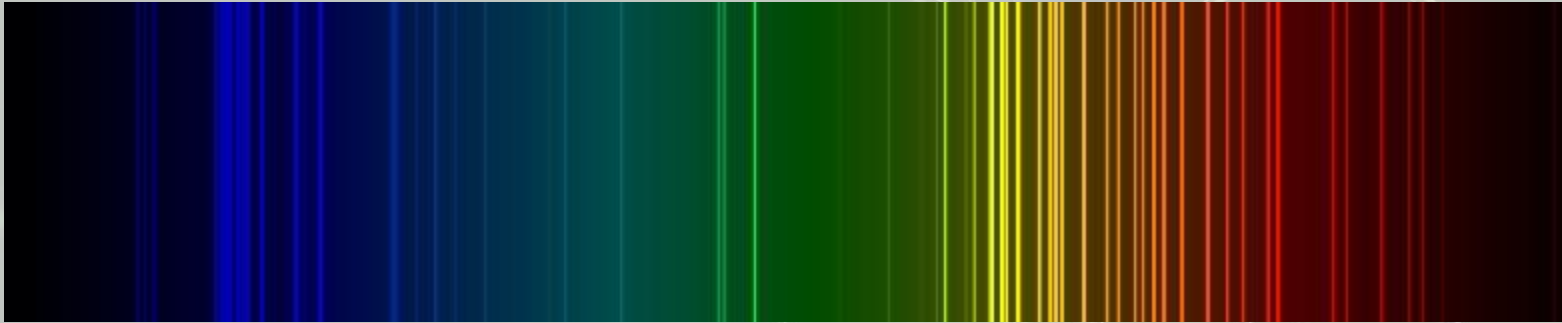
Absorption Spectrum



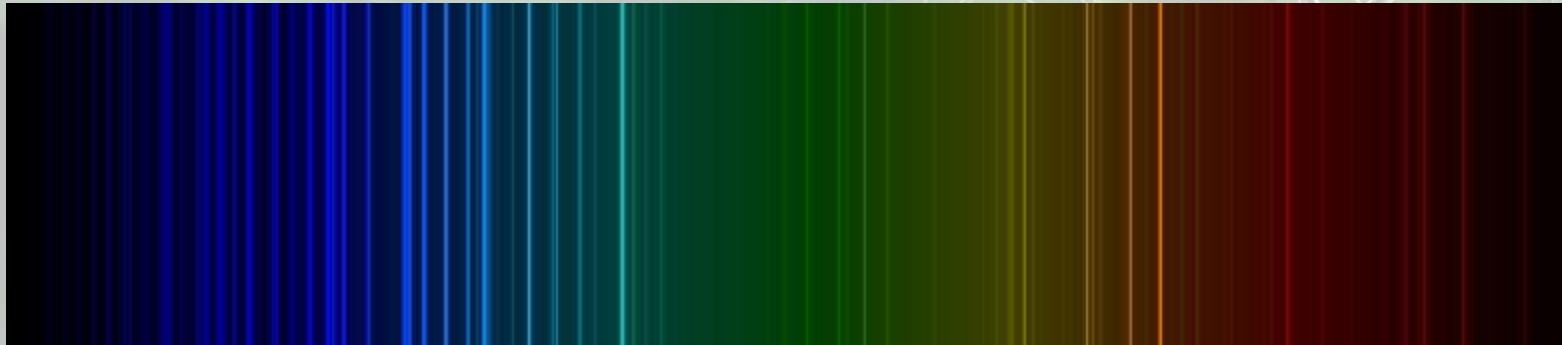
Below is a picture of the spectral lines given off by hydrogen. Note there are 3 different frequencies.



- Neon



- Argon



Practice

- Calculate the wavelength and energy of a photon that moves from the 1s orbital to the $2p^1$ orbital in an atom with a frequency of 2.9×10^{15} Hz.
- How much energy would a mole of these photons have in kJ/mol?

Those exciting electrons!

- We just learned that electrons could be excited from their *ground state* to a higher level *excited state* by the absorption of light.
- The excited electrons can then release energy (in the form of light) and return to their ground state.
- The amounts of energy absorbed or released are very predictable because the energies of electrons in orbit are very predictable.

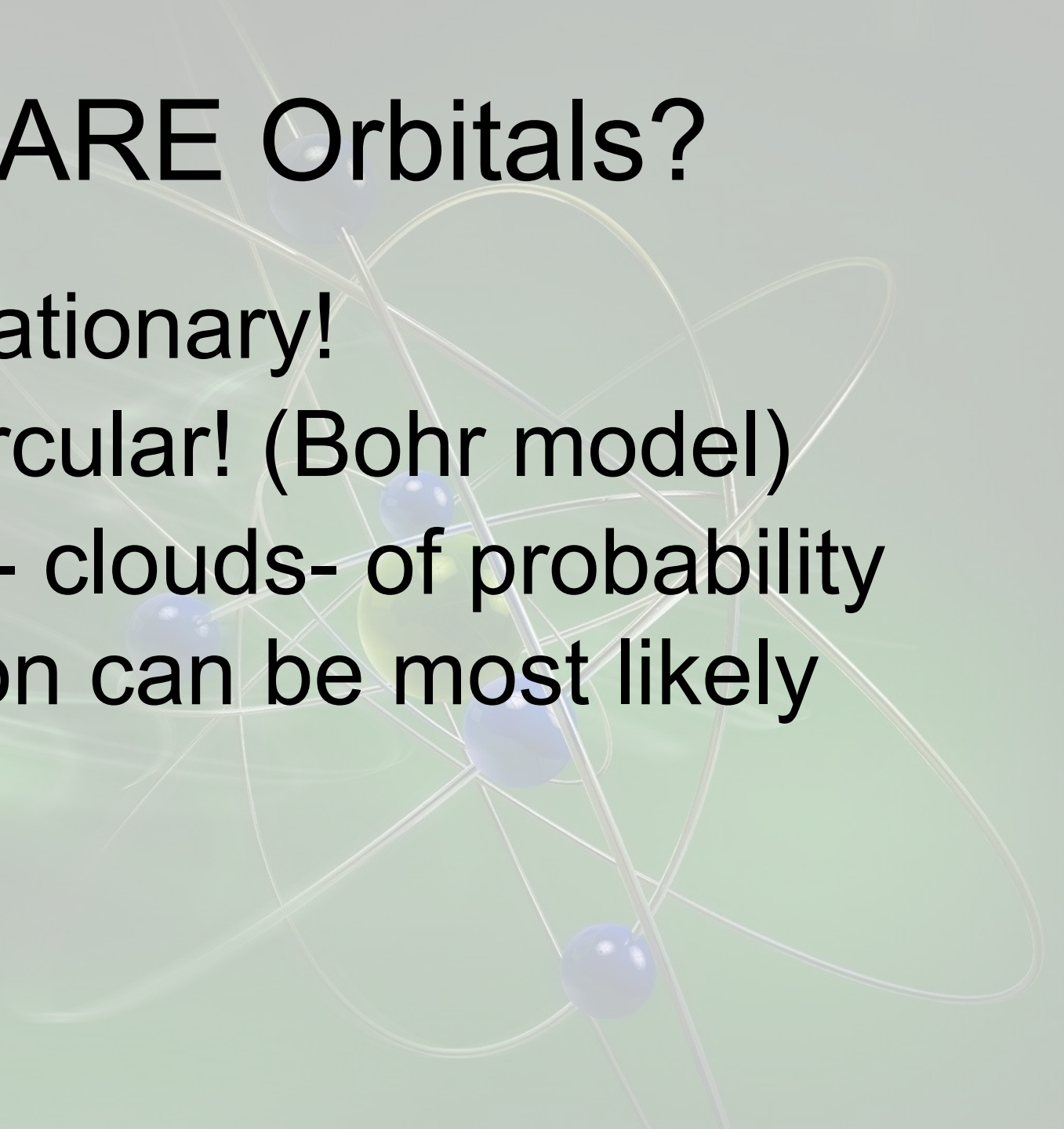
What did energy show about the atom:

Orbital diagrams

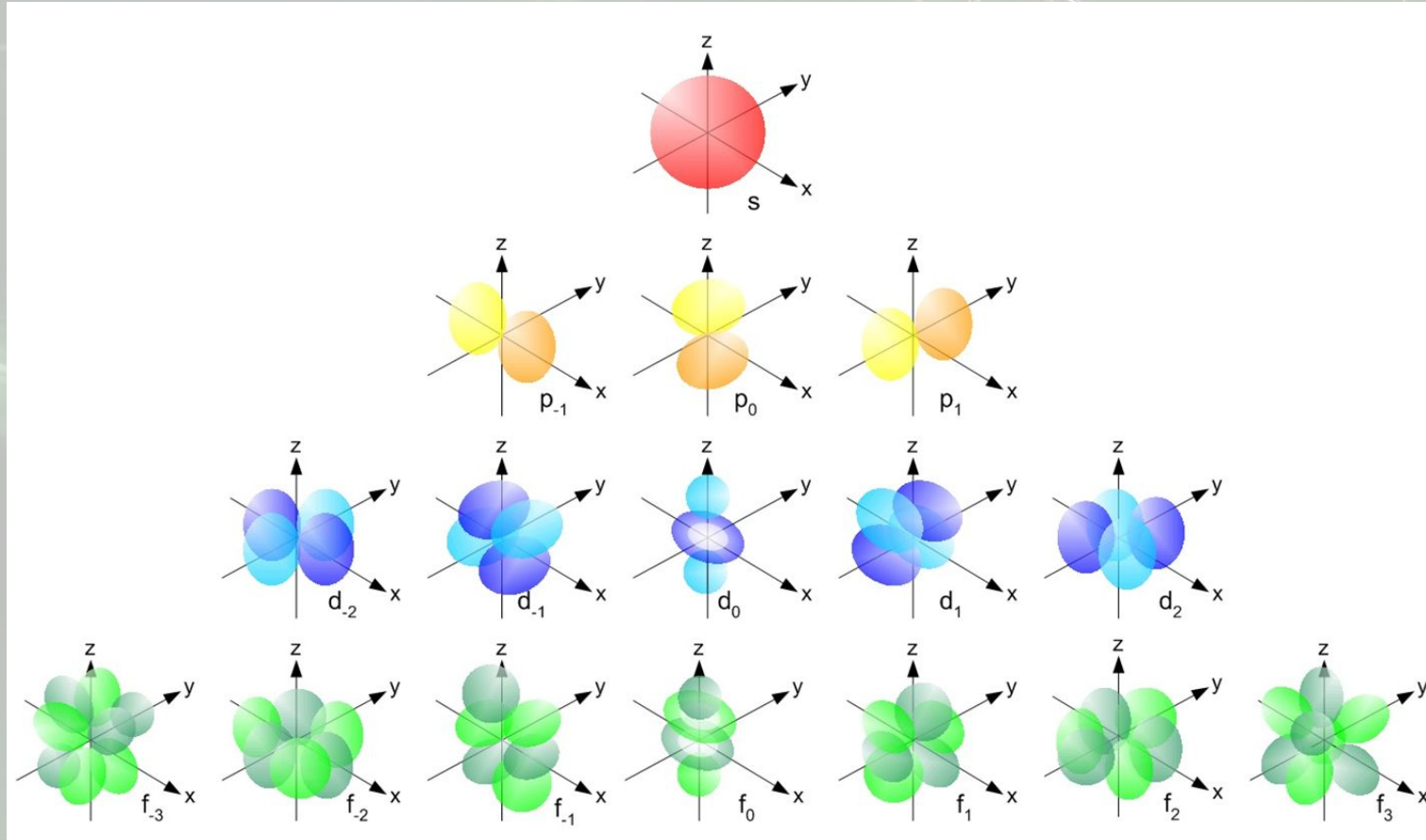
- Electrons orbit around the nucleus in nonrandom patterns called 'orbits'.
- Each atom has many orbits of differing energies. The lower the energy, the closer to the nucleus. Each atom has a unique set of energies.
- There are rules for filling orbitals with electrons.

But what ARE Orbitals?

- . They are NOT stationary!
- . They are NOT circular! (Bohr model)
- . They are spaces- clouds- of probability where the electron can be most likely found



Electrons are arranged in the atom in a series of levels and sublevels (shells and subshells).



Electron Configurations

- The ways in which electrons are arranged in various orbitals around the nuclei of atoms are called **electron configurations**.
 - Three rules—the Aufbau principle, the Pauli exclusion principle, and Hund's rule—tell you how to find the electron configurations of atoms.

Electron Configurations

- **Aufbau Principle**

- According to the aufbau principle, electrons occupy the orbitals of lowest energy first

- **Pauli Exclusion Principle**

- According to the Pauli exclusion principle, an atomic orbital may describe at most two electrons. To occupy the same orbital, two electrons must have opposite spins (two seats in a loveseat!)

- **Hund's Rule**

- Hund's rule states that electrons occupy orbitals of the same energy in a way that makes the number of electrons with the same spin direction as large as possible. (No unnecessary loveseat sharing!)

The Periodic Table is arranged according to electron configuration... read it like a book

Diagram illustrating the periodic table arranged by electron configuration, showing the relationship between Period, Group, and the corresponding subshells.

Periods (Rows): 1, 2, 3, 4, 5, 6, 7

Groups (Columns): 1A, 2A, 3A, 4A, 5A, 6A, 7A, 8A

Subshells:

- s-block (Purple):** 1s, 2s, 3s, 4s, 5s, 6s, 7s
- p-block (Yellow):** 2p, 3p, 4p, 5p, 6p
- d-block (Blue):** 3d, 4d, 5d, 6d
- f-block (Grey):** 4f, 5f

Labels: La (Lanthanum), Ac (Actinium)

Let's Try Some!

- Na

- C

- Xe

- S⁻²



Orbital Diagrams

- Orbital diagrams show the electrons as half arrows in a representation of their locations.
- If the electron configuration is like the ROSTER (list) of electrons in the atom, then the orbital diagram is like a SEATING CHART
- Rules- fill lowest levels first, NO SHARING unless they have to, all same spin until sharing is forced

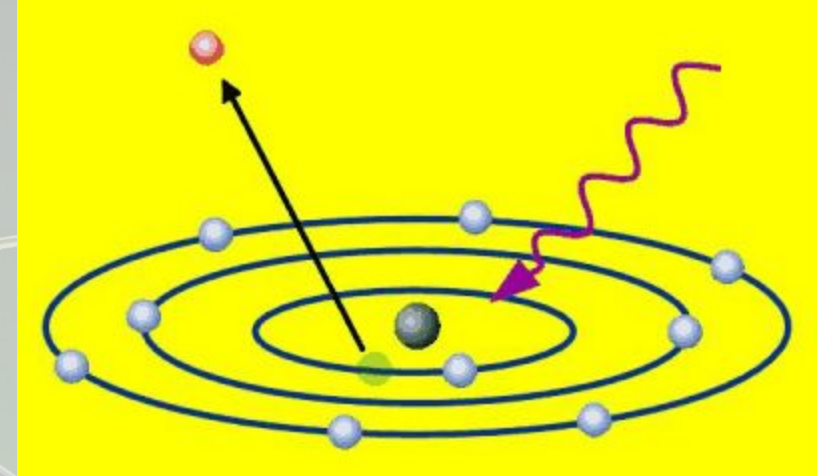
Table 5.3

Electron Configurations for Some Selected Elements

Element	Orbital filling						Electron configuration
	1s	2s	2p _x	2p _y	2p _z	3s	
H	↑						1s ¹
He	↑↓						1s ²
Li	↑↓	↑					1s ² 2s ¹
C	↑↓	↑↓	↑	↑			1s ² 2s ² 2p ²
N	↑↓	↑↓	↑	↑	↑		1s ² 2s ² 2p ³
O	↑↓	↑↓	↑↓	↑	↑		1s ² 2s ² 2p ⁴
F	↑↓	↑↓	↑↓	↑↓	↑		1s ² 2s ² 2p ⁵
Ne	↑↓	↑↓	↑↓	↑↓	↑↓		1s ² 2s ² 2p ⁶
Na	↑↓	↑↓	↑↓	↑↓	↑↓	↑	1s ² 2s ² 2p ⁶ 3s ¹

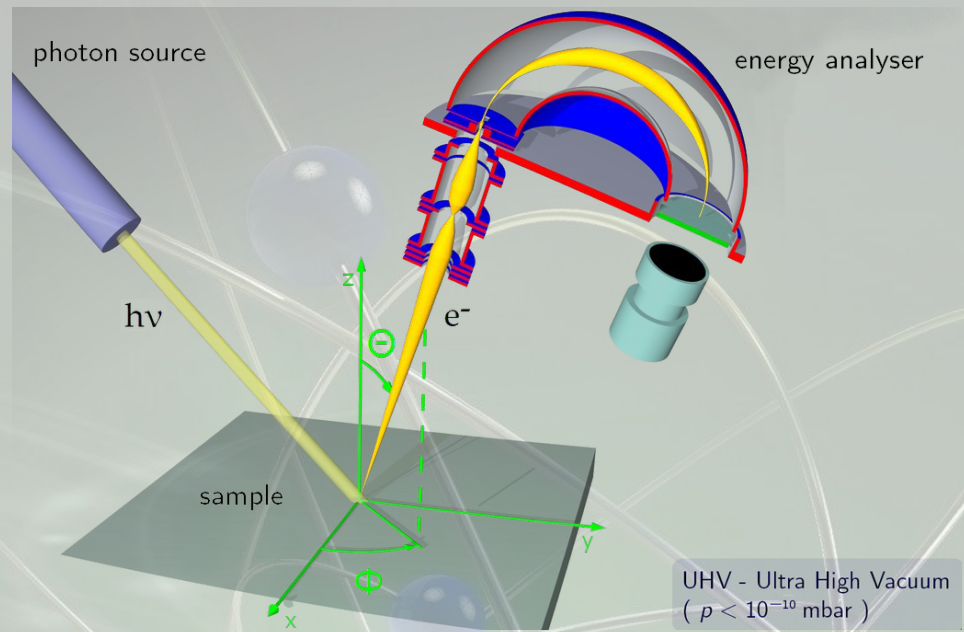
**Unit 7: Atomic Theory
and Periodicity**
Part 3: Photoelectron Spectroscopy

Photoelectron Spectroscopy (PES)



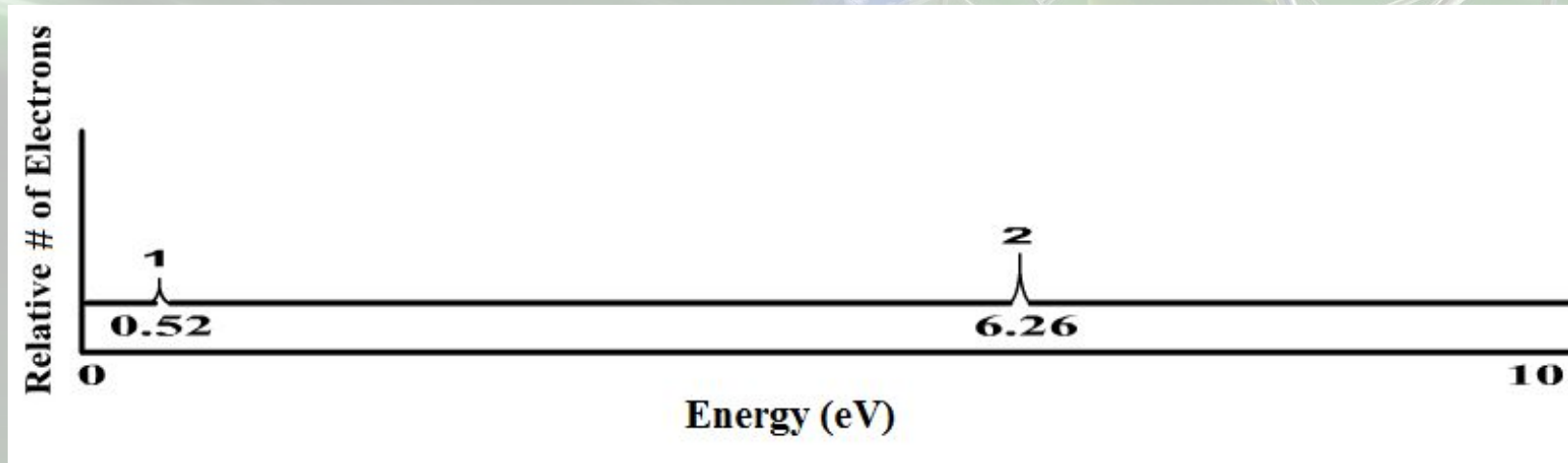
-uses the energy from electrons emitted via the photoelectric effect to gain information about the electronic structure of a substance. Very high energy photons of light of a given wavelength shine on a gas phase sample of the element. If the photons are higher in energy than the ionization of the electrons, one electron per atom is ejected.

Since there are a limited number of electrons (and energy levels) in an atom, there are a limited number of energies of electrons emitted. These energies can be measured. When the kinetic energy of the emitted electron is subtracted from the energy of the light photons, the resulting value is the ionization energy of that particular electron.

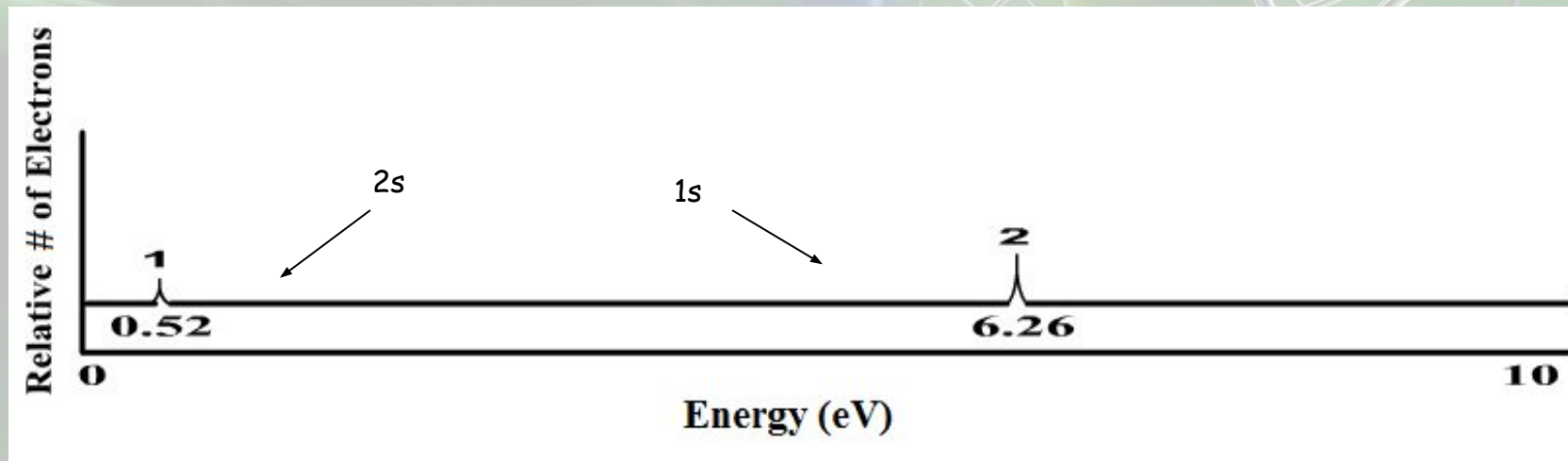


The photons used for photoelectron spectroscopy range from ultraviolet light to X-rays. Analysis of photoelectron spectra for a single element can give us information about the shells and subshells of electrons and the number of electrons in each.

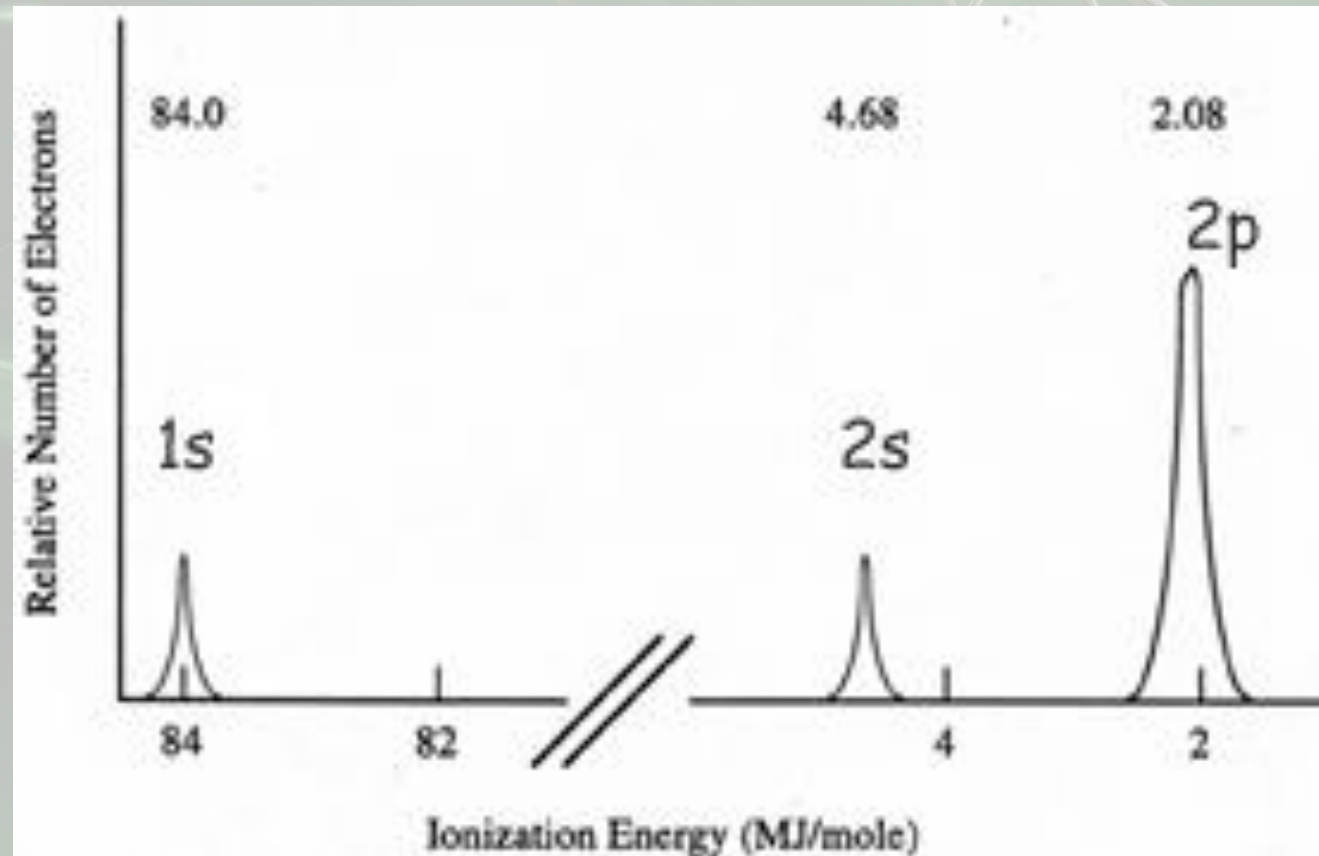
The x-axis has units of ionization energy of the electron. Electrons closer to the nucleus have greater Coulombic attraction and thus have a higher ionization energy. Some scales show the energies going from highest to lowest and some go from lowest to highest, so attention needs to be paid to these numbers. Don't worry about the units, because they may vary. The height of the peak indicates the number of electrons in the atom having that energy (the number of electrons in a sublevel).



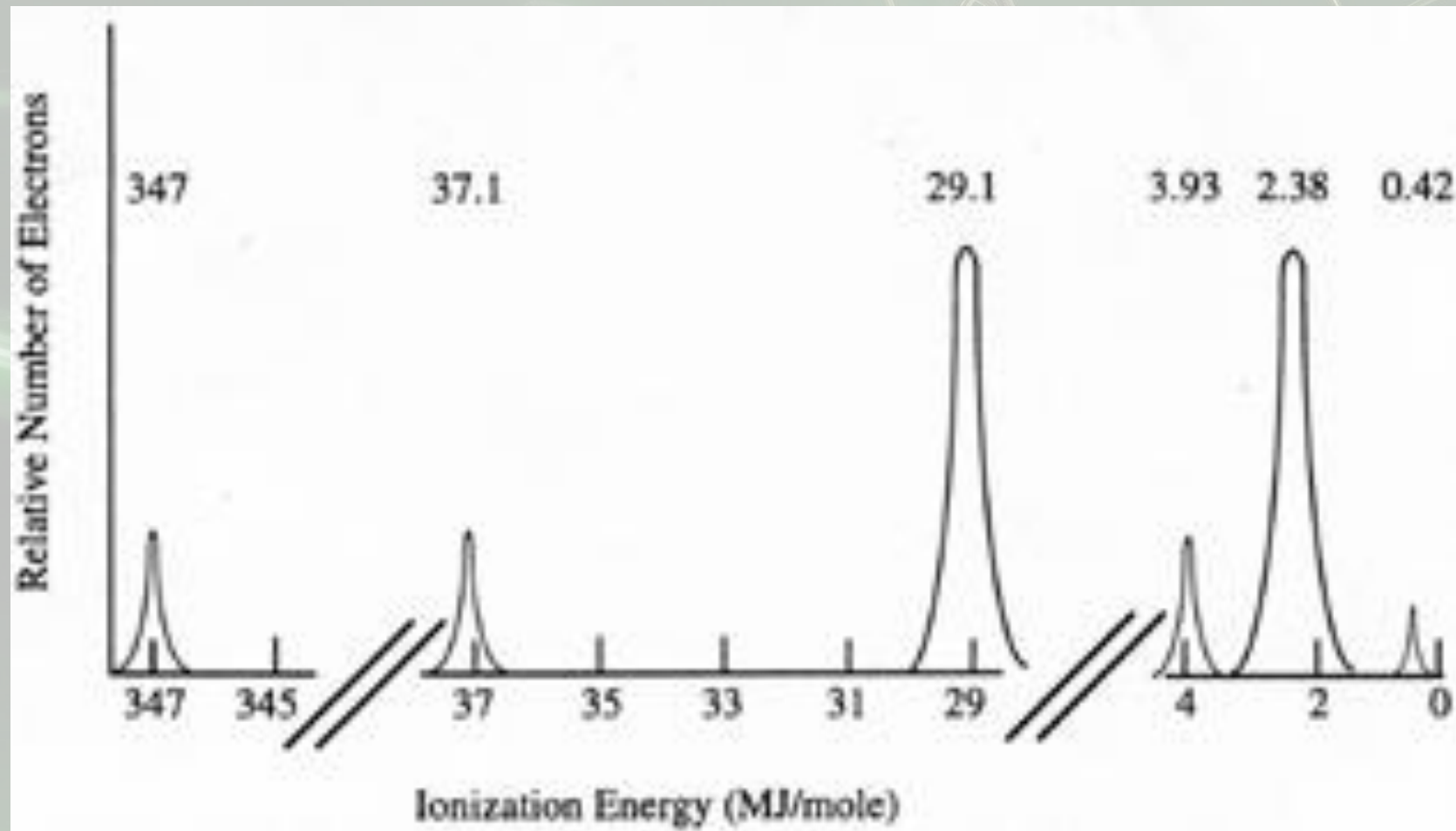
The graph below shows a simulated photoelectron spectrum for the element lithium. What is the electron configuration of lithium? Notice that the energy goes from lowest to highest from left to right on this chart. The 2nd peak is twice as large as the first peak. Remember that electrons that are lower in energy must be farther from the nucleus.



Notice that the energy goes from highest to lowest on this graph.



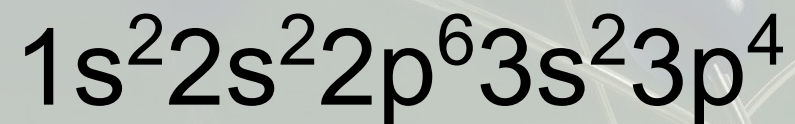
What element is this?



Day 2 Warm Up

- Magnesium atoms emit photons of a wavelength of 192 nm in their emission spectra. What is the frequency of these photons?
- What is the energy of these photons?
- This amount of energy is for ONE photon from ONE atom's emission. How would you calculate the amount of energy per mole of magnesium? (express your answer in kJ/mol)

WARM UP!



Atoms of an element, X, have the electronic configuration shown above. The compound most likely formed with aluminum.

- (A) Al_2X
- (B) AlX_2
- (C) Al_2X_3
- (D) Al_3X_2

Matching!

(A) $1s^2 2s^2 2p^5 3s^2 3p^5$

(B) $1s^2 2s^2 2p^6 3s^2 3p^6$

(C) $1s^2 2s^2 2p^6 2d^{10} 3s^2 3p^6$

(D) $1s^2 2s^2 2p^6 3s^2 3p^5$

(E) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$

1. An impossible electron configuration
2. The ground-state configuration for the atoms of a transition element
3. The ground-state configuration of a of a halogen
4. The ground-state configuration of a common ion of an alkaline earth element
5. An excited atom

Exceptions to Aufbau principle:

Cu and **Cr** are two elements that have exceptional electron configurations.

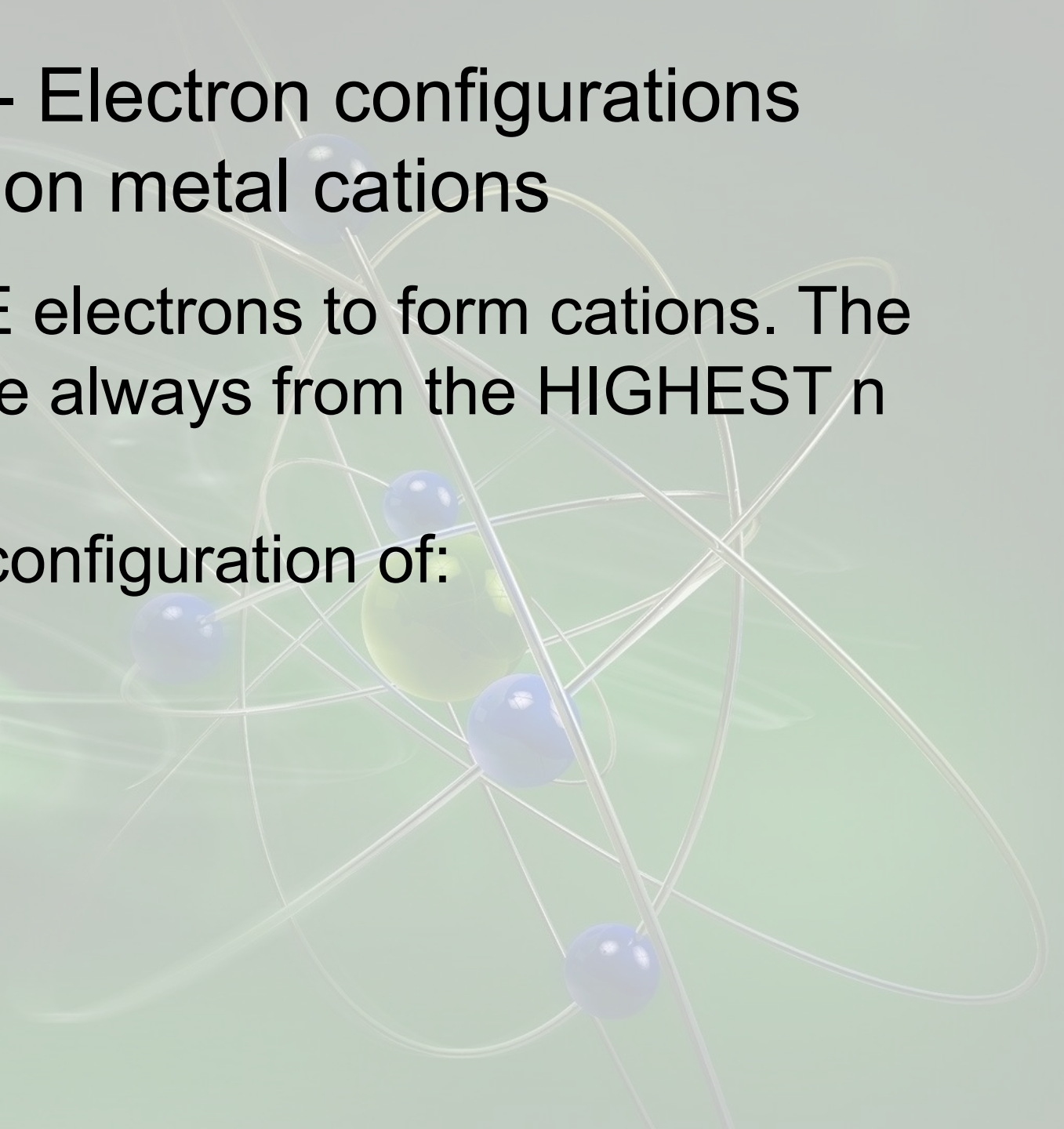
- Cu has a configuration that ends in $4s^1 3d^{10}$ instead of $4s^2 3d^9$.
- Cr is $4s^1 3d^5$ instead of $4s^2 3d^4$.
- Because fewer electrons are sharing orbitals, electron repulsion is decreased and stability is increased.
- *** this is what I call a '5 point'. Want a 5? Know it. Want a 3? Don't worry about it.

These exceptions do not need to be memorized for the AP test, but if given the exceptions, you should be able to explain why.

K $4s^1$	Ca $4s^2$	Sc $3d^1$	Ti $3d^2$	V $3d^3$	Cr $4s^1 3d^5$	Mn $3d^5$	Fe $3d^6$	Co $3d^7$	Ni $3d^8$	Cu $4s^1 3d^{10}$	Zn $3d^{10}$	Ga $4p^1$	Ge $4p^2$	As $4p^3$	Se $4p^4$	Br $4p^5$	Kr $4p^6$

Another '5 point'- Electron configurations of transition metal cations

- Transition metals LOSE electrons to form cations. The FIRST electrons lost are always from the HIGHEST n level shell.
- Example: the electron configuration of:
 - Fe^{+2}
 - Cu^{+2}





Unit 7: Atomic Theory and Periodicity

Part 4: Periodic Table Trends

Periodic Trends

- Periodic Trends are specific properties that are present in the periodic table that illustrate different aspects of a certain element.
- Almost all of the properties that are asked about in exam questions rely on Coulombic (electrostatic) attraction between outer electrons and the nucleus. Hence, you should refer to Coulomb's law in your answers!

$$E \propto \frac{Q_1 Q_2}{r}$$



THE ANSWER IS COULOMB'S LAW.

WHAT'S THE QUESTION?

made with mematic

Coulombs Law

$$E \propto \frac{Q_1 Q_2}{r}$$

- E = ionization energy, the energy needed to remove the outermost electron.
- Q_1 = charge of an electron, -1.
- Q_2 = effective nuclear charge (Z_{eff}) of unshielded protons in nucleus.
- r = distance between charged particles, which can be approximated by the period (n, energy level).



In short, the energy of attraction or repulsion between charged particles is:

- directly proportional to the **magnitude** (size) of the charges
- inversely proportional to the **distance** between the charges

All Periodic Trends can be understood in terms of Coulomb's Law(!)

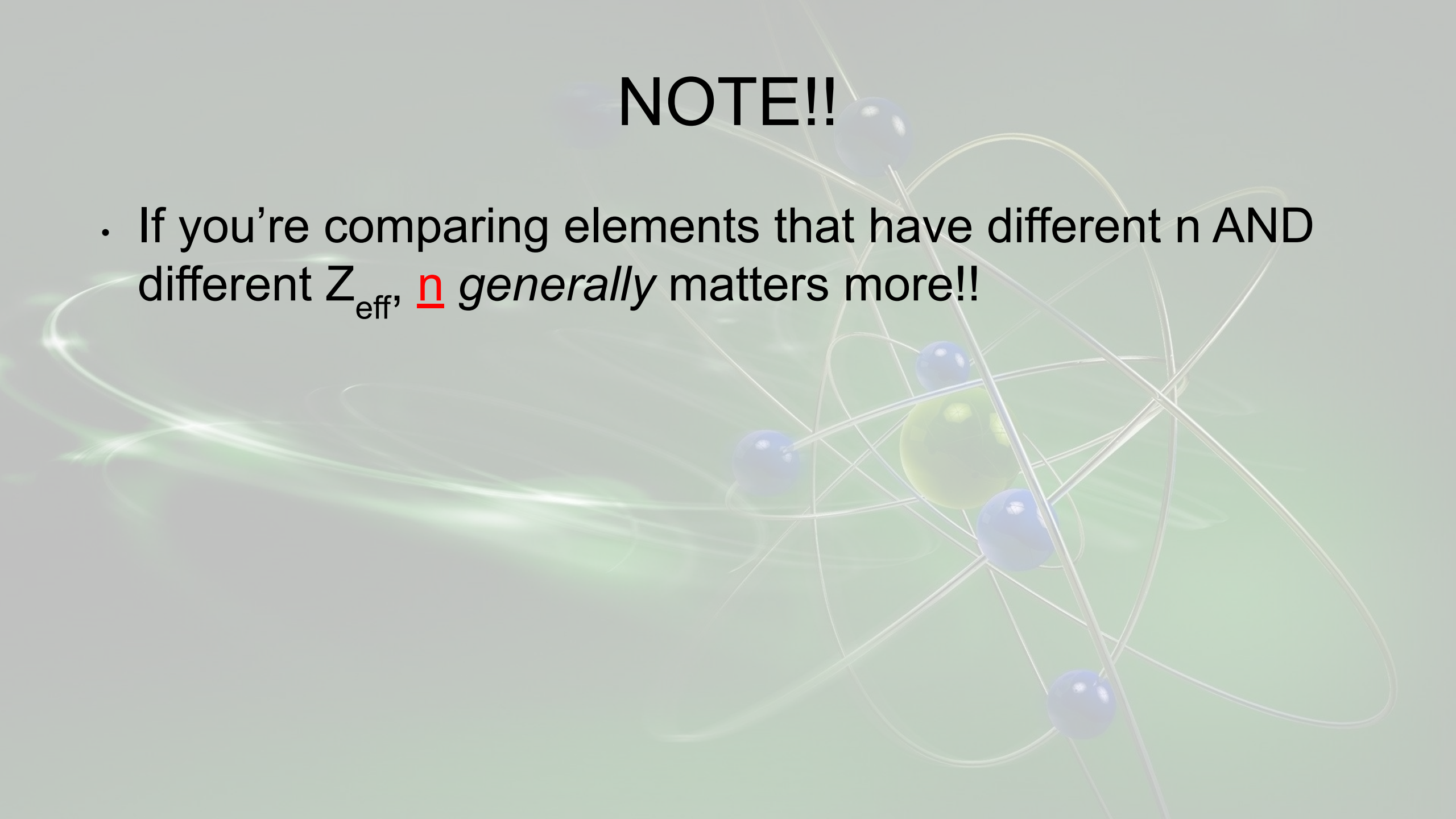
- Electrons are attracted to the protons in the nucleus of an atom.
 - The closer an electron is to the nucleus, the **MORE** strongly it is attracted.
 - Distance between electrons and the nucleus can be approximated by n (main energy level)
 - The more (unshielded) protons in a nucleus, the **MORE** strongly an electron is attracted.
- Electrons are repelled by other electrons in an atom.
 - If other electrons are between a valence electron and the nucleus, the valence electron will be **LESS** strongly attracted to the nucleus (this is known as shielding).

Effective Nuclear Charge (Z_{eff}): net positive charge experienced by electrons

- Attraction of a valence electron to a proton is partially shielded by the inner shell (core) electrons, and so the farther away an electron is from the nucleus, the less + charge it feels from the nucleus!
- Thus, following Coulomb's Law:
- The nucleus of atoms with a **higher Z_{eff}** (at the same energy level, n) will be more attractive to their valence electrons
- The nucleus of atoms with a **lower Z_{eff}** (at the same energy level, n) will be less attractive to their valence electrons
- At the same energy level, Z_{eff} is directly proportional to the number of valence e^- !

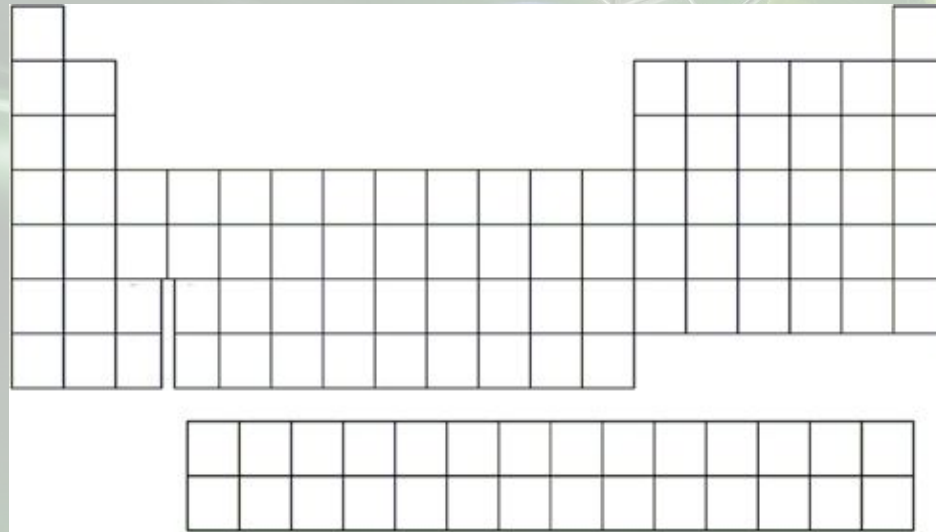
NOTE!!

- If you're comparing elements that have different n AND different Z_{eff} , n *generally* matters more!!



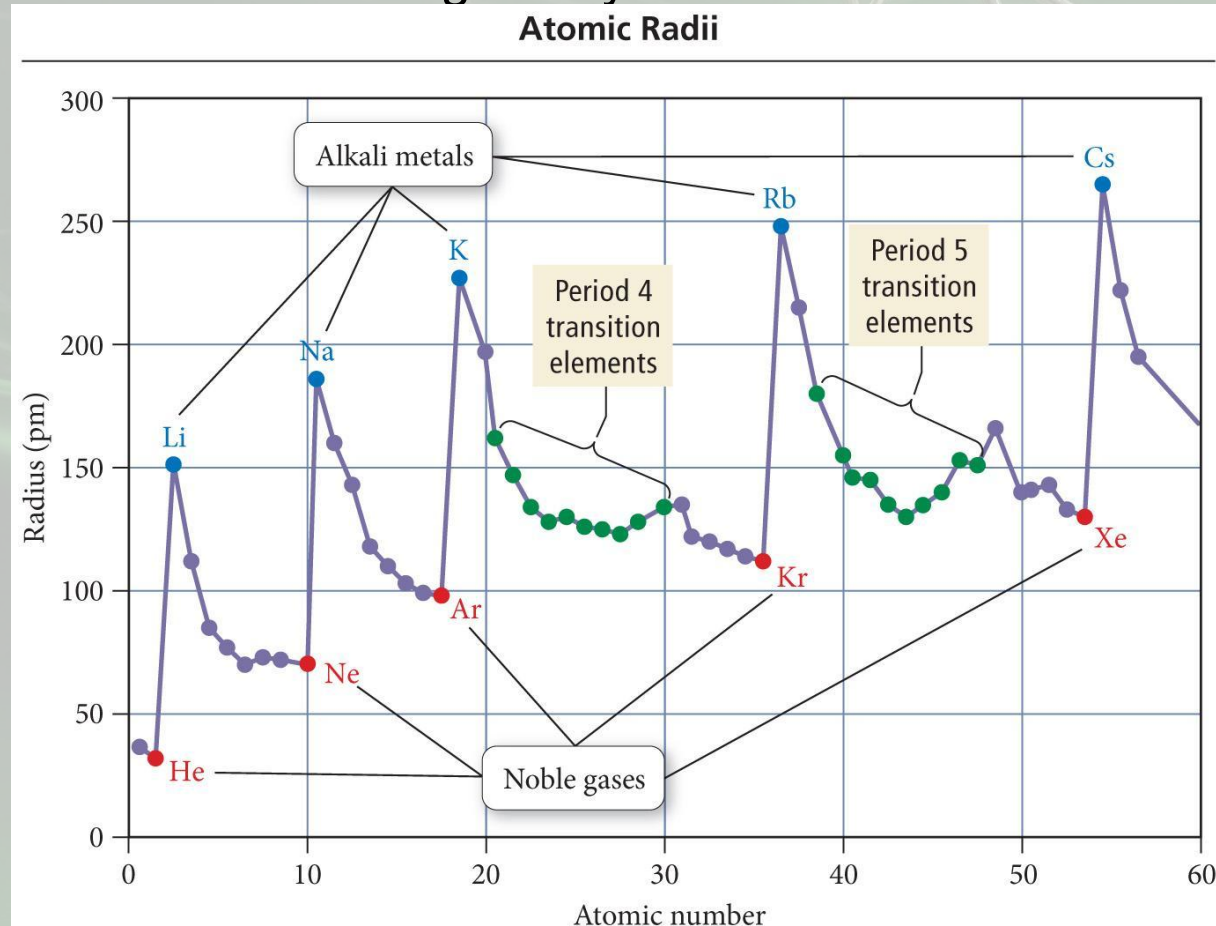
Trendy Trends: Periodic Trends to Know

- **Atomic radius** (size of atom): distance between the nucleus and valence electrons.



Trend in Atomic Radius – Main Group

- Atomic radius increases down group
 - ✓ Valence shell farther from nucleus
 - ✓ Effective nuclear charge fairly close



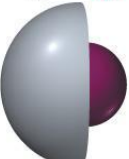
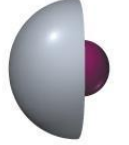
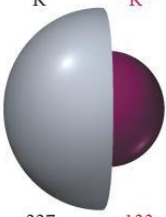
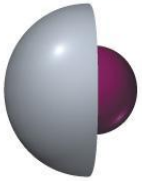
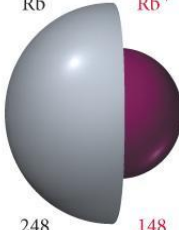
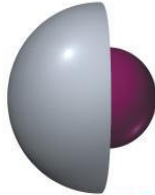
Ionic radius: distance from the nucleus to valence electrons in a charged atom.

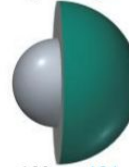
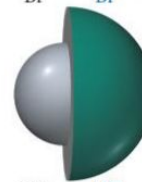
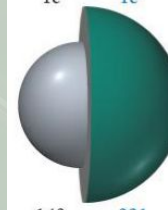
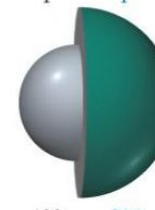
- *Most* comparisons between an atom and their ion or ions of the same atom can be explained by **e^-/e^- repulsion**:
 - Positively charged cations are **SMALLER** than the neutral atom because missing electrons in the outermost shell results in less e^-/e^- repulsion, thus valence electrons are closer to the nucleus.
 - Negatively charged anions are **LARGER** than the neutral atom because more electrons in the outermost shell results in more e^-/e^- repulsion, thus valence electrons are farther from the nucleus.

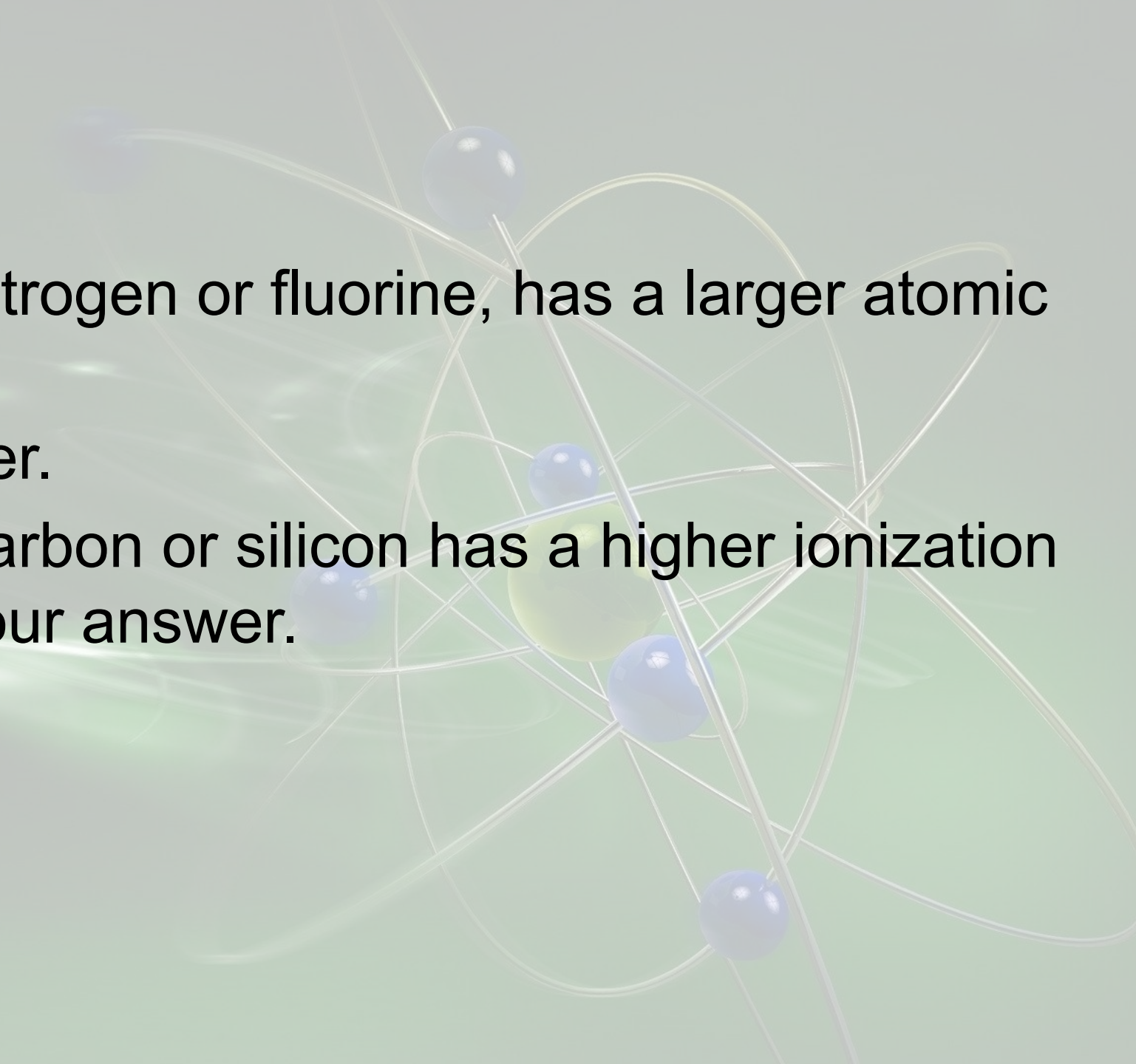
Only one type of comparison between an atom and their ion(s) should **NOT** utilize the e^-/e^- repulsion argument:

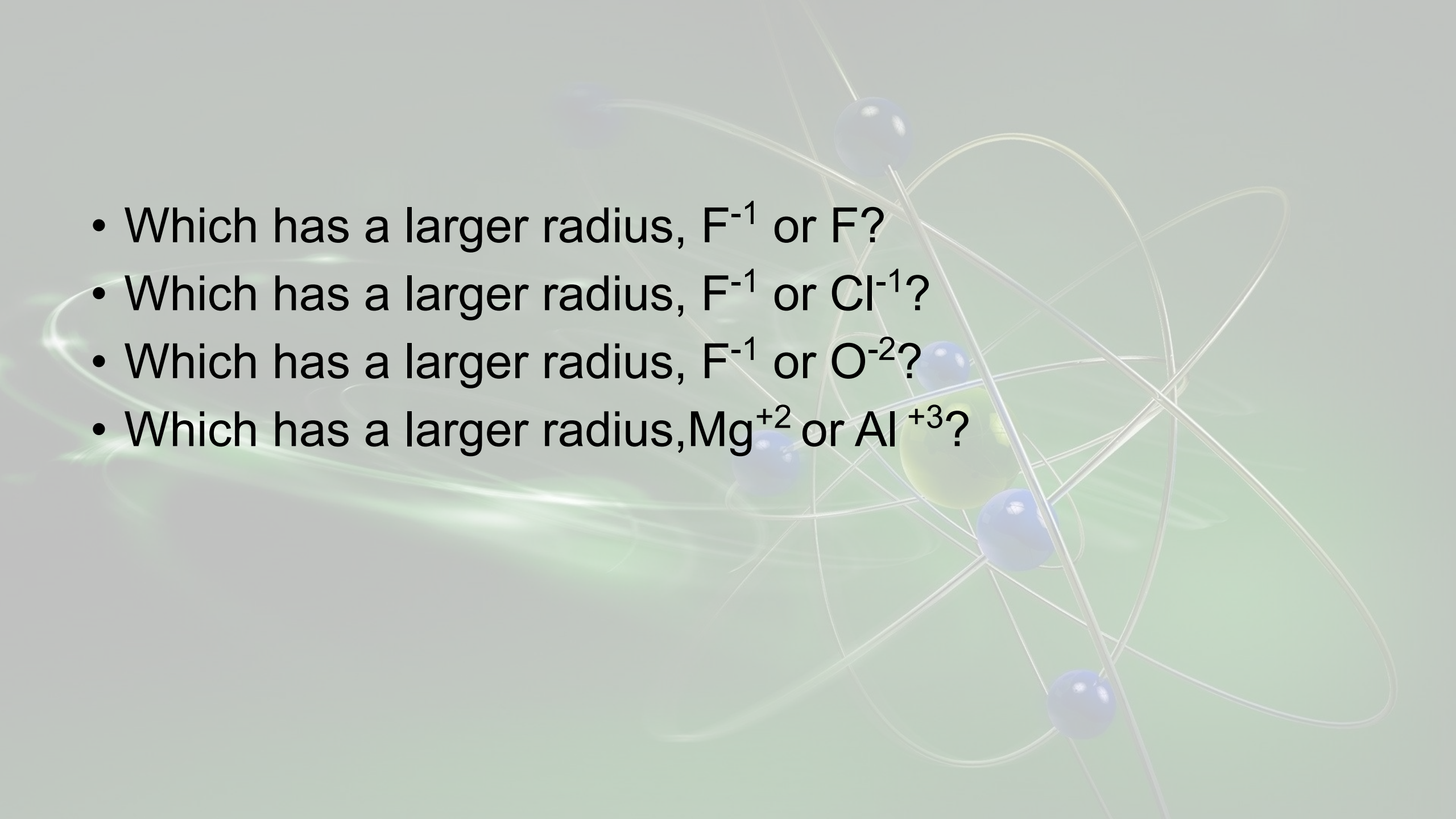
- Metals cations which have lost sufficient electrons such that their valence electrons are now in a lower energy level (n). Examples: Sr vs Sr^{2+} , Na vs Na^+ , Al^{2+} vs Al^{3+} , etc.
- Why? Because **n matters more than Z_{eff} !!!** If a species has their outermost electrons on a lower energy level (n), their valence electrons are closer to and thus more attracted to the nucleus.

Periodic Trends in Ionic Radius

Radii of Atoms and Their Cations (pm)		
Group 1A	Group 2A	Group 3A
Li Li^+  152 60	Be Be^{2+}  112 31	B B^{3+}  85 23
Na Na^+  186 95	Mg Mg^{2+}  160 65	Al Al^{3+}  143 50
K K^+  227 133	Ca Ca^{2+}  197 99	Ga Ga^{3+}  135 62
Rb Rb^+  248 148	Sr Sr^{2+}  215 113	In In^{3+}  166 81

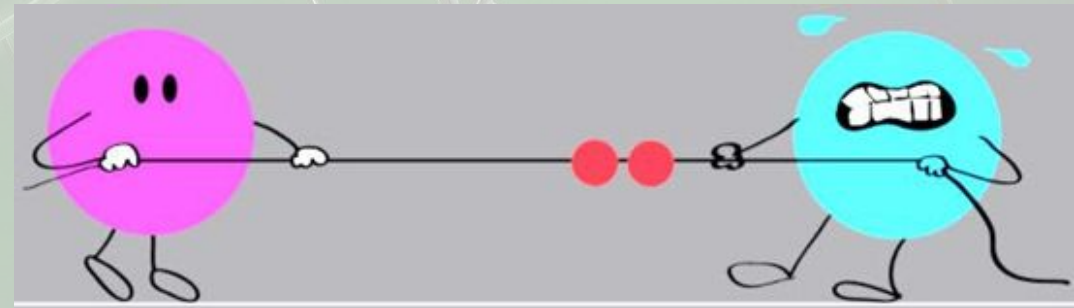
Radii of Atoms and Their Anions (pm)	
Group 6A	Group 7A
O O^{2-}  73 140	F F^-  72 136
S S^{2-}  103 184	Cl Cl^-  99 181
Se Se^{2-}  117 198	Br Br^-  114 195
Te Te^{2-}  143 221	I I^-  133 216

- 
- Which element, nitrogen or fluorine, has a larger atomic radius?
 - Justify your answer.
 - Which element, carbon or silicon has a higher ionization energy? Justify your answer.

- 
- Which has a larger radius, F^{-1} or F ?
 - Which has a larger radius, F^{-1} or Cl^{-1} ?
 - Which has a larger radius, F^{-1} or O^{-2} ?
 - Which has a larger radius, Mg^{+2} or Al^{+3} ?

Electronegativity

- **Electronegativity**: attraction of an atom for pair of valence level electrons in a covalent bond with another atom.
- Think of the atoms as playing “tug of war” with their valence shell electrons!
- Electronegativity of an atom is determined by combining the following two trends: **Ionization Energy vs Electron Affinity**



Ionization Energy vs Electron Affinity

- Ionization Energy
 - Energy required to remove an electron
 - $X(g) + \text{energy} \rightarrow X^+(g) + e^-$
 - endothermic (+kJ/mol)
(**always**)
 - **Higher** attraction between nucleus and electron = higher ionization energy and higher electron affinity!
- Electron Affinity
 - Energy change when electron is added to an atom
 - $X(g) + e^- \rightarrow X^-(g) + \text{energy}$
 - first electron added is always exothermic (−kJ/mol) (**usually**)

Metallic vs Non-Metallic Character

- Metallic Character
 - Metals react by losing electrons
 - How easy it is to remove an electron from an atom or ion
 - **Low** Ionization Energy = **High** Metallic Character
- Non-Metallic Character
 - Non-metals react by gaining electrons
 - How hard is it to remove an electron from an atom or ion
 - **High** Ionization Energy = **High** Non-Metallic Character

Practice

O Oxygen	F Fluorine
S Sulfur	Cl Chlorine

- Consider the elements shown above. Which element has the:
- Highest ionization energy? c) Highest Metallic Character?
- Highest electron affinity? d) Highest Non-Metallic Character?

Successive Ionization Energies

Removing one electron, and then another electron, and then another electron...

- 1st Ionization Energy: (IE_1) energy required to remove the first (highest energy) electron
- 2nd Ionization Energy: (IE_2) energy required to remove the second electron (second highest energy)

- Each additional electron requires **MORE** energy to remove than the previous one:

$$\text{IE}_1 < \text{IE}_2 < \text{IE}_3 \text{ (etc.)}$$

- Valence (outer) electrons require much **LESS** energy to remove than core (inner) electrons

$$\text{IE}_{\text{valence}} \ll \text{IE}_{\text{core}}$$

You can identify an element from the pattern of successive ionization energies!

- Examine the trend between the energy for the removal of valence versus core electrons. What do you notice?

TABLE 8.1 Successive Values of Ionization Energies for the Elements Sodium through Argon (kJ/mol)							
Element	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	IE ₇
Na	496	4560					
Mg	738	1450	7730				
Al	578	1820	2750	11,600			
Si	786	1580	3230	4360	16,100		
P	1012	1900	2910	4960	6270	22,200	
S	1000	2250	3360	4560	7010	8500	27,100
Cl	1251	2300	3820	5160	6540	9460	11,000
Ar	1521	2670	3930	5770	7240	8780	12,000

FRQ Practice

IE_1	IE_2	IE_3	IE_4	IE_5
801 kJ/mol	2,426 kJ/mol	3,660 kJ/mol	24,682 kJ/mol	32,508 kJ/mol

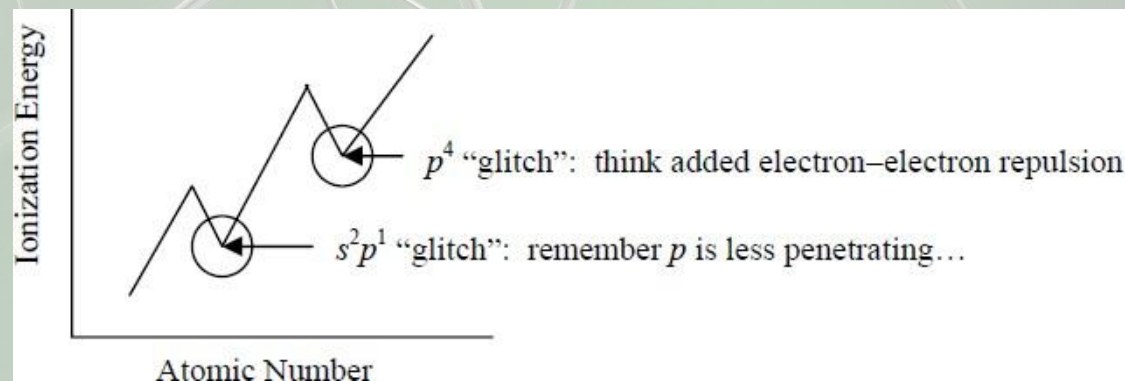
- Which element from Period 2 does the table of successive ionizations energies above represent? Explain.
- Would it require more energy to remove an electron from an F^+ ion or an F^{2+} ion? Justify your answer using Coulomb's Law.

Ionization Energy Exceptions:

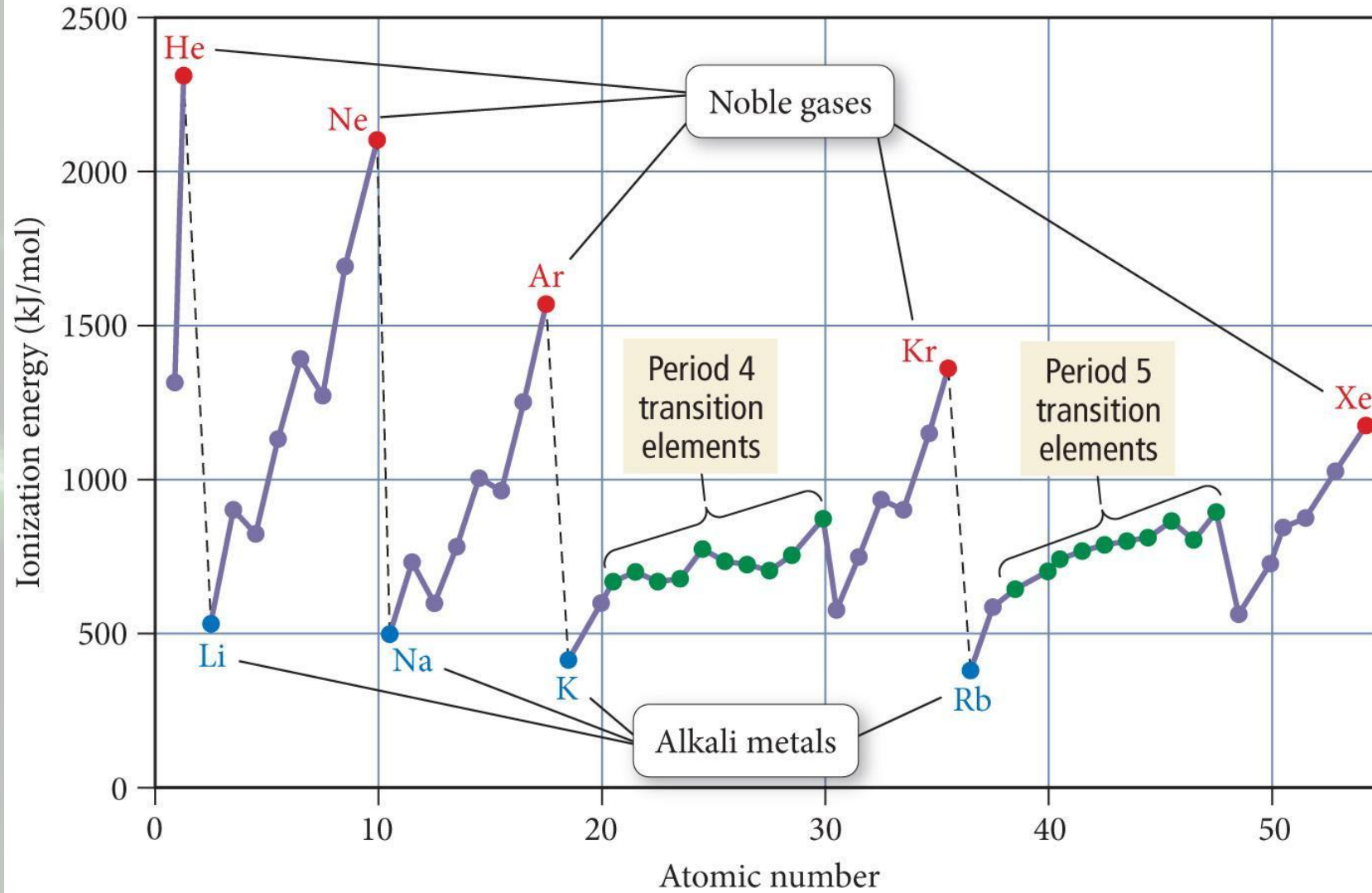
You MUST know these!

***these exceptions only matter when both elements are found on the same period (energy level).

- **Group 5 vs Group 6**: “ **p^4 glitch**” (Example: oxygen vs nitrogen): It’s easier to remove an electron from oxygen ($2p^4$) because of the **repulsion** of the paired electrons (whereas the $2p^3$ electrons in nitrogen (Group 5) are all unpaired).
-
- **Group 2 vs Group 3**: “ **s^2p^1 glitch**” (Example: beryllium vs boron): It’s easier to remove an electron from boron ($2s^22p^1$) because it’s being removed from the p orbital, and p orbitals do not **penetrate** the nuclear region as greatly as the s orbitals, so electrons in a p orbital are not as tightly held.



First Ionization Energies



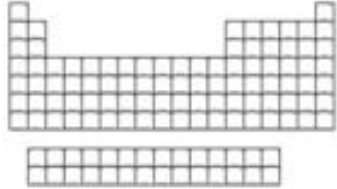
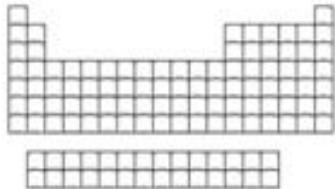
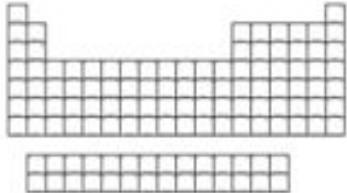
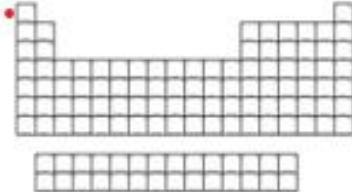
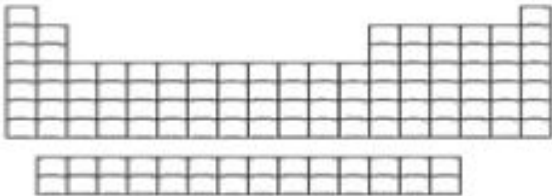
A large grid of 100 squares arranged in 10 rows and 10 columns. The grid is composed of several connected rectangular blocks. The top row has 10 squares. The second row has 10 squares. The third row has 10 squares. The fourth row has 10 squares. The fifth row has 10 squares. The sixth row has 10 squares. The seventh row has 10 squares. The eighth row has 10 squares. The ninth row has 10 squares. The tenth row has 10 squares. The grid is a single connected shape with no gaps between the blocks.

One last Trend: overall reactivity

- **Reactivity** depends on whether the element reacts by losing electrons (metals) or gaining electrons (nonmetals).
 - Metals are MORE reactive as you move down a column: because metals lose electrons as they react, **LESS** attraction between valence electrons and the nucleus will result in a more reactive metal.
 - Non-metals are LESS reactive as you move down a column: because non-metals gain electrons as they react, **LESS** attraction between valence electrons and the nucleus will result in a less reactive non-metal.

In Summary

Draw arrows in the direction that each trend increases!

Atomic Radius Trend 	Metallic Character Trend 
Electronegativity Trend 	Ionization Energy Trend 
Reactivity Trend 	

How to Answer Periodic Trends Free Response Questions

- Justifying all of the trends on the periodic table can be simplified using these two generalizations:
 - Use *number of protons* (or Z_{eff}) to justify trends across a period.
 - Use increased distance (greater value of n) to justify trends down a group.

How to Earn Full Points on Periodic Trends Problems

Follow these three steps EVERY time you answer a periodicity question!

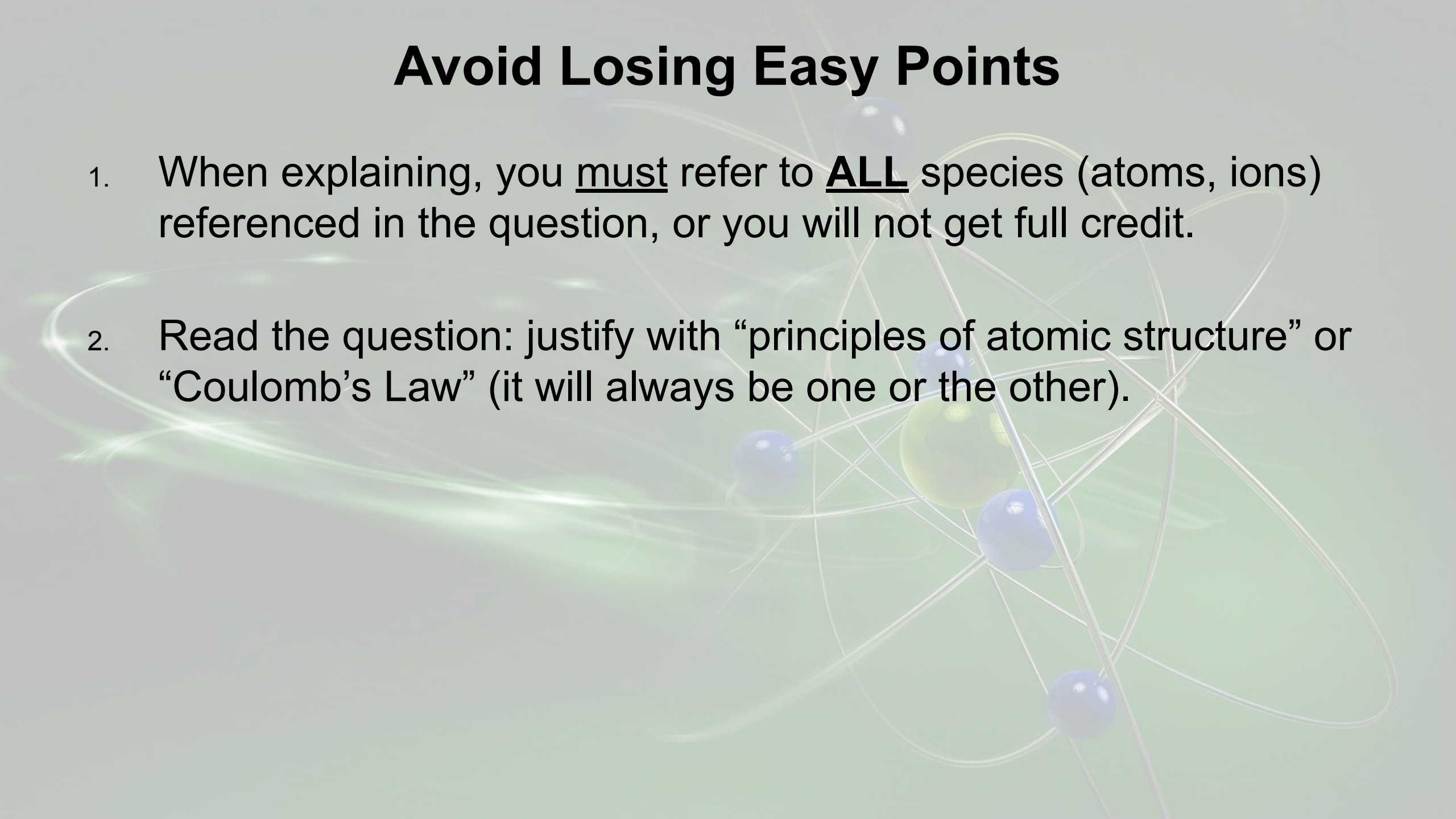
1. Locate *both* elements on the periodic table and **state** the principal energy level (n) and the sublevel containing the valence electrons for *each* element.
2. Do they have the same or different n values?
3. If same n : argue with **number of protons** if different n : argue with n vs. n **distance**.

REMEMBER: a trend is not an explanation!

Simply identifying a trend (atomic radius decreases as you move from left to right across a period, electronegativity decreases as you move down a column, etc) earns 0 points!

Avoid Losing Easy Points

1. When explaining, you must refer to **ALL** species (atoms, ions) referenced in the question, or you will not get full credit.
2. Read the question: justify with “principles of atomic structure” or “Coulomb’s Law” (it will always be one or the other).



Specific Question Types

1. Comparisons between isoelectronic species: explain with number of p^+

- Isoelectronic species with more protons are **SMALLER** because the valence electrons are more attracted to and thus **CLOSER** to the nucleus.
- Isoelectronic species with less protons are **LARGER** because the valence electrons are less attracted to and thus **FARTHER** from the nucleus.
- Example: Which has the larger radius, Mg^{+2} or Al^{+3} ? Justify.

Specific Question Types

2. Comparisons between an atom and its ion/ions of the same atom, same n: explain with **e^-/e^- repulsion**

- Positively charged cations are **SMALLER** than the neutral atom because of lower e^-/e^- repulsion, thus valence electrons are **CLOSER** to the nucleus.
- Negatively charged anions are **LARGER** than the neutral atom because of larger e^-/e^- repulsion, thus valence electrons are **FARTHER** from the nucleus.

Example: Which has the larger radius, Fe^{+2} or Fe^{+3} ? Justify.

Specific Question Types

- Comparisons between an atom and its ion/ions of the same atom, different n: explain with **distance**
 - If a species has their outermost electrons on a lower energy level (n), their valence electrons are closer to and thus more attracted to the nucleus.

Example: Which has the larger radius, Na^+ or Na ? Justify.

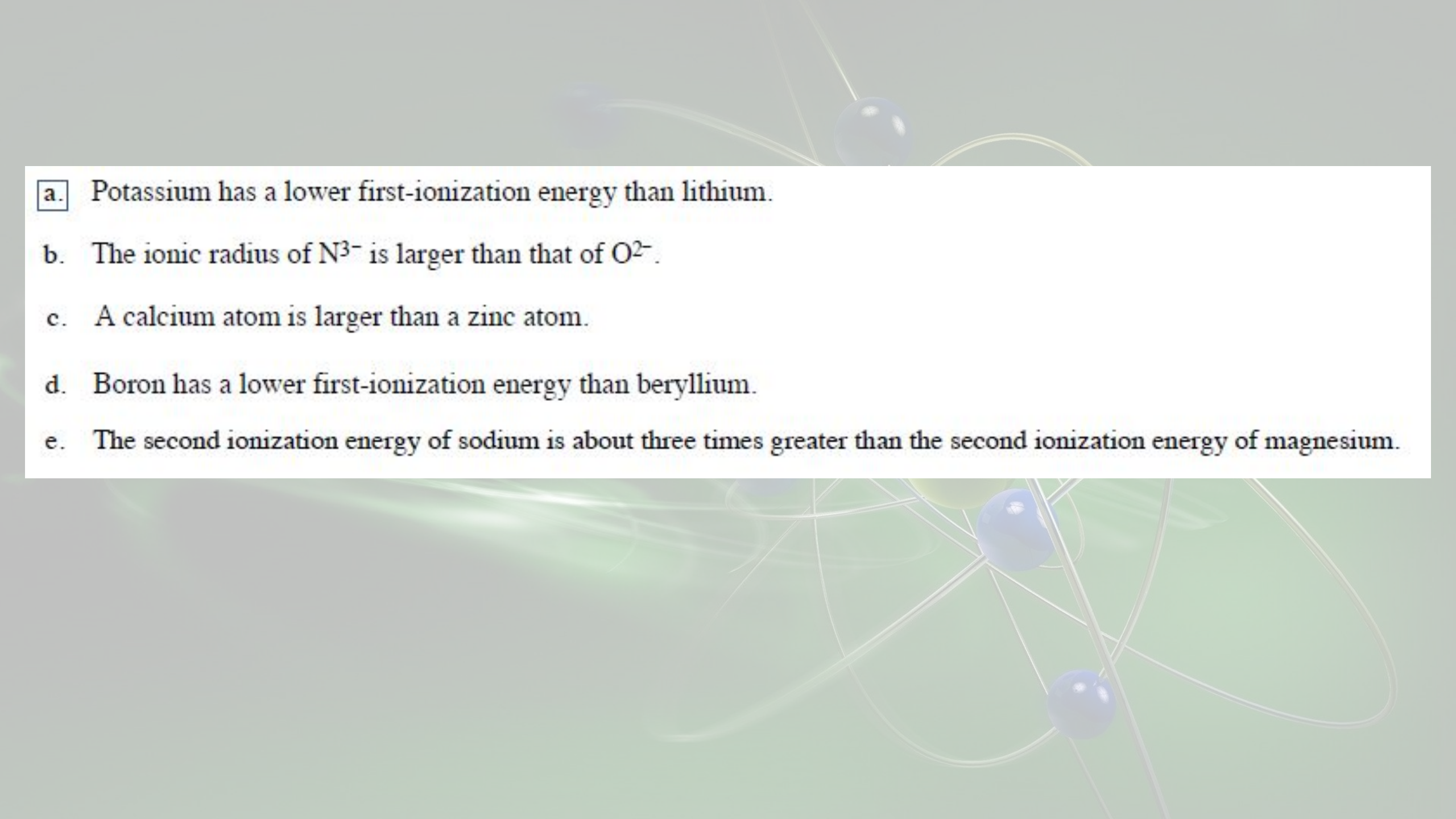
Guided Practice: How will an answer be different when explained in terms of “atomic structure” vs “Coulomb’s Law”?

Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+} ? Explain using principles of atomic structure.

Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+} ? Be sure to reference Coulomb’s Law in your explanation.

Periodic Trend Haikus

- a) Potassium has a lower first-ionization energy than lithium.
- b) The ionic radius of N^{3-} is larger than that of O^{2-} .
- c) A calcium atom is larger than a zinc atom.
- d) Boron has a lower first-ionization energy than beryllium.
- e) The second ionization energy of sodium is about three times greater than the second ionization energy of magnesium.

- 
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- d. Boron has a lower first-ionization energy than beryllium.
- e. The second ionization energy of sodium is about three times greater than the second ionization energy of magnesium.